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ATAD2 mediates chromatin-bound histone chaperone turnover

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eLife Assessment

This **valuable** study explores the role of the chromatin regulator ATAD2 in mouse spermatogenesis. The data **convincingly** demonstrate that ATAD2 is essential for proper chromatin remodeling in haploid spermatids, influencing gene accessibility, H3.3-mediated transcription, and histone eviction. Using *Atad2* knockout (KO) mice, the authors link ATAD2 to the DNA-replication-independent incorporation of sperm-specific proteins like protamines and histone H3.3. Although the findings highlight chromatin abnormalities and impaired in vitro fertilization in KO mice, natural fertility remains unaffected, suggesting possible in vivo compensatory mechanisms. Future experiments will be needed to tease out the precise molecular role of ATAD2 in spermatogenesis. This work will be of interest to the epigenetics and developmental fields.

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Abstract

ATAD2, a conserved protein which is predominantly expressed in embryonic stem (ES) cells and spermatogenic cells, emerges as a crucial regulator of chromatin plasticity. Our previous parallel studies conducted in both ES cells and *S. pombe* highlighted the fundamental role of ATAD2 in facilitating chromatin-bound histone chaperone turnover. Focusing on spermatogenesis, we demonstrate here that ATAD2 regulates the HIRA-dependent localization of H3.3 on the genome and influences H3.3-mediated gene transcription. Moreover, by modulating histone eviction and the assembly of protamines, ATAD2 ensures proper chromatin condensation and genome packaging in mature sperm. Disruption of *Atad2* function in mice leads to abnormal genome organization in mature spermatozoa. Together, these findings establish a previously overlooked level of chromatin dynamic regulation, governed by ATAD2-controlled histone chaperones binding to chromatin, which defines the balance between histone deposition and removal.

Introduction

Spermatogenesis involves unique and dramatic chromatin remodelling and genome reorganization events that take place in preparation for a shift from the universal nucleosome-based eukaryotic genome organization into a unique nucleoprotamine-based structure (Gaucher et al. 2010 [↗](#); Bao and Bedford 2016 [↗](#); Le Blevet et al. 2020 [↗](#); Okada 2022 [↗](#)). During the last decade, important information on the molecular basis of the nucleosome-to-nucleoprotamine transition during the postmeiotic phases of spermatogenesis became available. Several testis-specific H2A, H2B and H3 histone variants replace the canonical histones, locally or globally, at different stages, starting early during the commitment of cells into meiotic and post-meiotic differentiation (Hoghoughi et al. 2018 [↗](#)). Most of this nucleosome dismantling and assembly occurs in the absence of DNA replication. In other words, the replication-independent nucleosome assembly pathways, such as the well-documented HIRA-dependent H3.3 assembly, are expected to

be critical in ensuring the necessary chromatin remodeling in these cells. Finally, hints to the functions of transition proteins (TPs), and their functional relationship with protamines (PRMs) were proposed based on the study of a specific late-expressing H2A variant, H2A.L.2 (Barral et al. 2017b [\[1\]](#)). In the proposed model we suggest that TPs do not directly replace histones, but rather could be loaded on open H2A.L.2-containing nucleosomes to control PRM assembly. Nucleosome-to-nucleoprotamine transition also includes a TP-dependent pre-PRM2 processing, histone displacement by PRMs and the structuration of the nucleoprotamine-based genome organization in the mature spermatozoa (Barral et al. 2017b [\[1\]](#); Rezaei-Gazik et al. 2022 [\[2\]](#)). In mice, two protamines, PRM1 and PRM2, replace histones. PRM2 is initially synthesized as a precursor protein, which undergoes sequential proteolytic processing at its N-terminal region during its incorporation, ultimately producing the mature, fully processed PRM2 (Arevalo et al. 2022 [\[3\]](#)).

Taking into account this unique dynamic of post-meiotic cells' nucleosomes, it is expected that these cells massively express generic factors, or those predominantly expressed in spermatogenic cells that act on chromatin, including histone acetylating machineries, acetylated histone readers and histone variants and their chaperones, all of which mediate histone and non-histone protein exchanges. More specifically, the replication-independent nature of nucleosome assembly, both in cells undergoing meiosis and in post-meiotic cells, strongly suggests the involvement of the chaperone / histone substrate couple, HIRA and H3.3.

An important part of this replication-independent histone assembly and transitions in the genome organizations occurs in the context of a massive histone H4 hyperacetylation in the haploid spermatogenic cells, spermatids, mediated by NUT-p300/CBP histone acetyltransferase complex (Shiota et al. 2018 [\[4\]](#); Rousseaux et al. 2022 [\[5\]](#)). Hyperacetylation of histone H4, especially, the double acetylation on H4K5K8 in these cells, leads to the binding of H4 by the first bromodomain of BRDT and to histone replacement (Moriniere et al. 2009 [\[6\]](#); Gaucher et al. 2012 [\[7\]](#); Goudarzi et al. 2016 [\[8\]](#); Shiota et al. 2018 [\[4\]](#)). Additionally, Histone H4 hyperacetylation, especially at K5, could also be recognized by another bromodomain-containing factor of unknown function in spermatogenic cells, ATAD2 (Caron et al. 2010 [\[9\]](#); Morozumi et al. 2016 [\[10\]](#)).

ATAD2 is a highly conserved bromodomain-containing factor that also harbours an AAA ATPase domain (Boussouar et al. 2013 [\[11\]](#); Cattaneo et al. 2014 [\[12\]](#)). Our early work showed that Atad2 is predominantly expressed in ES cells, as well as in spermatogenic cells, and is aberrantly activated in virtually all solid cancers (Caron et al. 2010 [\[9\]](#)). Although ATAD2's function in cancer prompted many investigations (Liu et al. 2022 [\[13\]](#)), its function in its physiological context of expression has remained largely overlooked. We reported the first investigations of ATAD2's function in ES cells (Morozumi et al. 2016 [\[10\]](#)). These studies in ES cells showed that ATAD2 is recruited into acetylated transcriptionally active chromatin and is critical to keep the chromatin of ES cells highly dynamic, as illustrated by an ATAD2-dependent high rate of histone exchange (Morozumi et al. 2016 [\[10\]](#)). Subsequent studies conducted by our laboratory, in ES cells and in *S. pombe*, deciphered the molecular basis of this activity of ATAD2, which relies on its capacity to ensure the turnover of the chromatin-bound histone chaperone, HIRA and FACT (Wang et al. 2021 [\[14\]](#)). In addition to ES cells, *Atad2* is also highly expressed in male germ cells, pointing to its potential role in the regulation of HIRA-dependent H3.3 assembly in these cells, especially, in post-meiotic cells. Indeed, previous published works have highlighted the important function of H3.3 assembly in meiotic and post-meiotic male genome programming (van der Heijden et al. 2007 [\[15\]](#); Yuen et al. 2014 [\[16\]](#); Tang et al. 2015 [\[17\]](#); Fontaine et al. 2022 [\[18\]](#)). We therefore hypothesized that ATAD2-regulated HIRA-H3.3 activity would also be a critical player in the post-meiotic male genome reorganization, as HIRA has been shown to be essential post-fertilization (Lin et al. 2014 [\[19\]](#)) and during oogenesis (Nashun et al. 2015 [\[20\]](#)).

Using a lacZ tagged *Atad2* knockout (KO) mouse model, we could show that *Atad2*'s expression is enhanced in post-meiotic male germ cells, ensuring a proper turnover of histone H3.3, including on the sex chromosomes. More remarkably, our data demonstrate that ATAD2 controls the appropriate expression of a series of genes whose activity is known to be H3.3-dependent in post-meiotic cells. Additionally, in the absence of ATAD2, perturbed nucleosome structural organization leads to a delayed histone-to-PRM-replacement, leading to a defective global final male genome

compaction. Although this defect in mature spermatozoon genome compaction has minimal effect on male fertility during natural reproduction, it impacts the success rate of *in vitro* fertilization using spermatozoa from *Atad2* KO mice pointing to a more fragile genome in these cells.

In summary, our work revealed a novel mechanism for the regulation of chromatin plasticity, involving the control of histone chaperone actions, particularly that of HIRA, controlling the proper male germ cell transcriptional regulation, genome compaction and optimal male fertility.

Results

***Atad2* is predominantly expressed in haploid male germ cells**

Atad2 is normally expressed in embryonic stem (ES) and spermatogenic cells and remains silent in the majority of adult tissues (Caron et al. 2010 [↗](#); Morozumi et al. 2016 [↗](#)). Therefore, in addition to ES cells, male germ cells constitute the second cell type where the molecular basis of the physiological ATAD2 activity could be investigated.

To decipher the function of ATAD2 during spermatogenesis, we used a mouse model harboring an insertion of a *lacZ* reporter cassette between exons 11 and 12 that enables splicing and fusion of the *lacZ* transgene with exon 11 of the long and short forms of *Atad2* (Fig. 1A [↗](#)) and production of a non-functional ATAD2-lacZ fusion protein (Fig. 1B [↗](#)), referred to as *Atad2* KO in this study. Furthermore, we crossed these mice with transgenic mice expressing ubiquitously Cre recombinase under CMV promoter, in order to remove the *neo* cassette and to delete *Atad2*'s exon 12 (Fig. 1B [↗](#)). In addition to the long somatic form of ATAD2, ATAD2L, the mouse testis encodes a short isoform with predominant expression in spermatogenic cells that we previously named ATAD2S (Caron et al. 2010 [↗](#)). The combined LacZ fusion and Cre recombinase-dependent deletion of *Atad2*'s exon 12 make both ATAD2L and ATAD2S undetectable with our homemade antibody, while as expected, a reduced level of both ATAD2 isoforms in *Atad2* heterozygous cells can be observed (Fig. 1C [↗](#)).

The expression of *lacZ* under the control of the *Atad2* promoter allows one to visualize *Atad2* gene activity in a stage-specific manner. X-gal staining of seminiferous tubule sections from *Atad2* KO mice showed a remarkably high expression of *Atad2* in post-meiotic spermatogenic cells (Fig. 1D [↗](#)). It is noteworthy that, as expected, no staining was observed in the wild-type tubules.

ATAD2 controls HIRA accumulation

In our previous work performed in both *S. pombe* and mouse ES cells, we demonstrated the crucial role for ATAD2 in HIRA-dependent histone dosage as well as in HIRA-chromatin interaction. Among other discoveries, this work showed that in the absence of ATAD2, HIRA becomes trapped on chromatin leading to its accumulation in ES cells, as well as in a variety of cancer cell lines (Wang et al. 2021 [↗](#)). Because of this very conserved action of ATAD2 in controlling the HIRA activity, we reasoned that the ATAD2-dependent HIRA activity should be particularly active during spermatogenic cell differentiation, especially, in post-meiotic cells, when ATAD2 expression is highly induced (Fig. 1D [↗](#)). To test if the absence of ATAD2 could lead to the accumulation of HIRA, as we previously reported in other systems, we quantified HIRA in spermatogenic cells from both wild-type and *Atad2* KO mice. These investigations revealed an accumulation of HIRA protein in *Atad2* KO spermatogenic cells (Fig. 2A [↗](#)), consistent with its increased association with chromatin (Wang et al. 2021 [↗](#)). We have also monitored HIRA accumulation as a function of spermatogenic cells stages. Due to the strong activation of *Atad2* after meiosis, in spermatids (Fig. 1D [↗](#)), the ATAD2-dependent accumulation of HIRA was expected to mostly occur in post-meiotic cells. Accordingly, we also compared HIRA accumulation in fractionated spermatogenic cell extracts. This experiment demonstrated that, in the absence of functional ATAD2, the accumulation of HIRA primarily occurs in round spermatids (Fig. 2B [↗](#)). This finding is consistent with the post-meiotic activation of *Atad2* and its role in regulating the turnover of chromatin-bound HIRA.

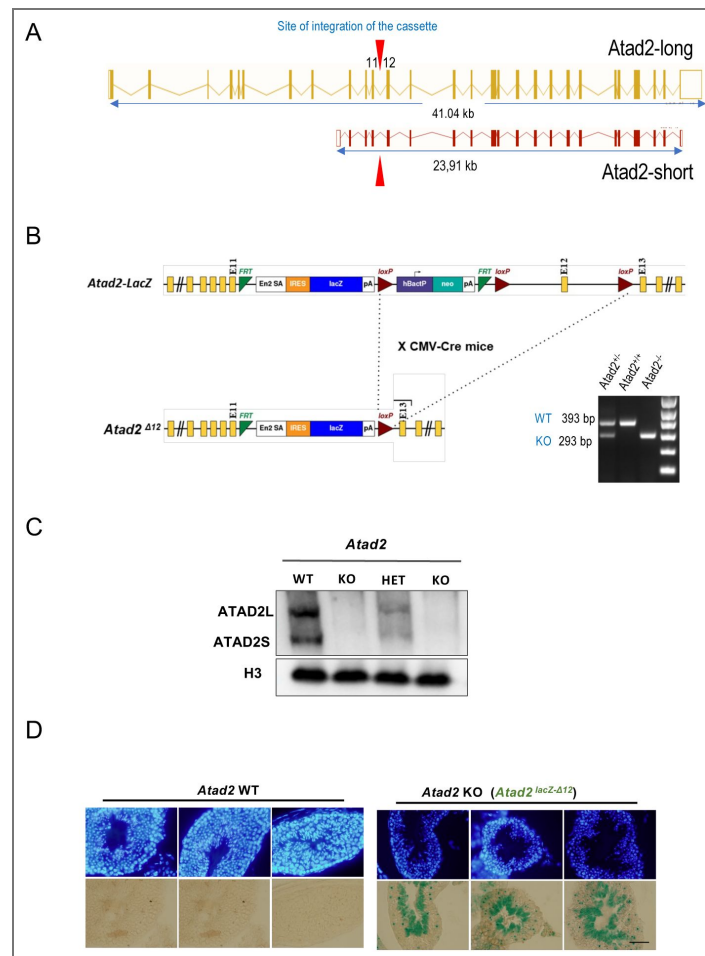


Figure 1. *Atad2* is highly expressed in post-meiotic male germ cells.

(A) The exon/intron organization of *Atad2* gene and the two ATAD2-encoding transcriptional units (encoding for ATAD2-long and ATAD2-short) are indicated (Dyer et al. 2025). The red arrows indicate the position of insertion of the *Lac-Z-Lox-Neo-Lox* cassette. (B) Gene targeting strategy for the generation of the *Atad2* KO allele is represented. Exon number and relative position, selection gene cassettes, LoxP sites and Frt sites are indicated. Crossing mice bearing this construct with *CMV-Cre* mice resulted in the generation of the *Atad2* LacZ^{Neo-Exon12} allele, which can be verified through PCR amplification of the genomic DNA (shown on the ethidium bromide-stained gel). The expected amplified bands for each genotype allowing the detection of both wild-type and KO alleles of *Atad2* are indicated. E = Exon. (C) Total protein extracts from wild-type (WT), heterozygous (HET) and *Atad2* knockout (KO) mice testes were probed with anti-ATAD2 and anti-H3 antibodies as indicated. Four different mice were used for this experiment. (D) Seminiferous tubules sections from wild-type and *Atad2* KO mice were stained with X-gal to visualize *lacZ* gene expression and β -galactosidase activity under the endogenous *Atad2* gene promoter. Testes from three different wild-type and *Atad2* KO mice were used to generate the represented sections. Scale bar: 200 μ m

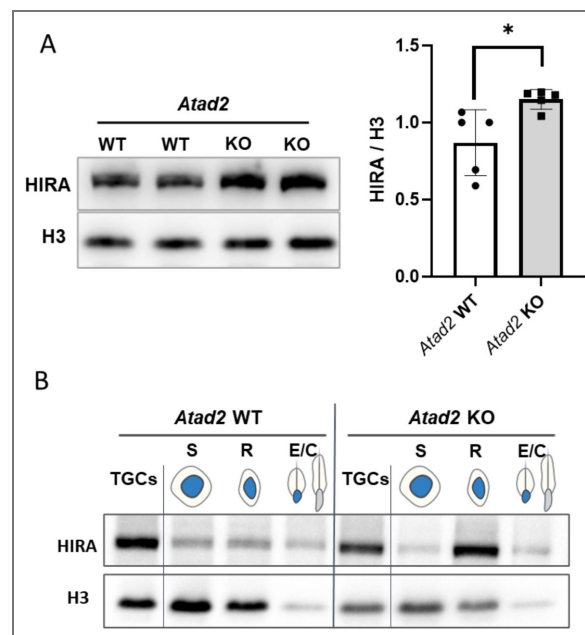


Figure 2. *Atad2* gene inactivation leads to the accumulation of HIRA.

(A) Total protein extracts from the testes of five wild-type (WT) and five *Atad2* KO mice were used to detect HIRA and H3 by immunoblotting. The panel shows a representative immunoblotting after HIRA detection. The histogram in the right panel shows the quantification of HIRA immunoblotting signal normalized to H3. The mean $\pm$ standard deviation was 0.86 ± 0.21 for wild type and 1.15 ± 0.06 for *Atad2* KO samples. The p value was 0.02 for unpaired Student t-test. (B) Protein extracts from fractionated spermatogenic cells from wild-type (*Atad2* WT) and knockout (*Atad2* KO) mice (pool of testes from 3 individuals per genotype) were used to detect HIRA and H3 by immunoblotting.

ATAD2 controls HIRA-dependent H3.3 localization

We speculated that an increased presence of HIRA on chromatin in the absence of ATAD2, should also be associated with increased amounts of deposited histones. Indeed, we have demonstrated previously in *S. pombe* that histone gene inactivation could rescue the growth arrest due to the absence of *Atad2/Abo1*. In parallel, we found that in ES cells, in the absence of ATAD2, an increased nucleosome assembly occurs, even on the so-called nucleosome-free region (NFR) (Wang et al. 2021 [DOI](#)).

Interestingly, visualization of H3.3, the major target of HIRA, by immunofluorescence, in *Atad2* KO round spermatids, displayed a diffuse, intense staining (Fig. 3 [DOI](#)), which is normally concentrated on the sex chromosomes (van der Heijden et al. 2007 [DOI](#); Fontaine et al. 2022 [DOI](#)) in the wild-type cells. Indeed, as expected and previously reported, in the wild-type round spermatids, H3.3 was mostly observed in a single domain, close to the unique chromocenter in these cells (Fig. 3 [DOI](#), left panels). In the absence of ATAD2, in agreement with an increased residence time of HIRA on chromatin in general and a reduced H3.3 turnover, an increased level of H3.3 could be observed on the sex chromosome (adjacent to the chromocenter), and more interestingly, on all over the genome (Fig. 3 [DOI](#), right panels).

General chromatin accessibility and transcriptional regulation by ATAD2

H3.3 is known to be dynamically incorporated on a group of active genes, especially around their transcriptional start sites (TSS) (de Dieuleveult et al. 2016 [DOI](#)). We therefore predicted that the observed increased H3.3 residence time in the absence of ATAD2 would decrease TSS regions' accessibility specially those of highly active genes. To verify this prediction, we purified round spermatid fractions from wild-type and *Atad2* KO mice testes and measured the general chromatin accessibility following an ATAC-seq approach. Chromatin accessibility was then assessed around the TSS of genes, which were categorized into four groups based on the quartile of their mean expression from three independent transcriptomes of wild-type round spermatids (Fig. 4A [DOI](#)), from the top 25% most highly expressed genes (group 1) to the 25% least expressed genes (group 4). As predicted, the absence of ATAD2 was associated with a significant decrease in TSS region accessibility of the most active genes (Fig. 4B [DOI](#) and 4C [DOI](#)), supporting a role for ATAD2 in maintaining chromatin dynamics at the most highly active chromatin regions.

To visualize the genes more sensitive to ATAD2-mediated chromatin remodeling that affect transcription, we monitored whole genome transcriptional activity during the first spermatogenic wave at day 20 post-partum (PP), when meiosis is complete, and in the first post-meiotic cells that appear on 22-, 24- and 26-days PP.

In order to definitively confirm our working hypothesis on the ATAD2-HIRA-H3.3 functional interaction, we took advantage of published H3.3-dependent transcriptomic data (Fontaine et al. 2022 [DOI](#)). Indeed, the authors of this publication identified a series of genes that are misregulated in cells with inactivated *H3.3b* gene. Because of the post-meiotic major activation of *Atad2*, we specifically focussed on the gene lists that show a H3.3-dependent deregulated expression in post-meiotic round spermatids.

In these transcriptomes, we specifically followed the expression of the published lists of genes that are downregulated or upregulated in the absence of H3.3 (Fontaine et al. 2022 [DOI](#)). Very interestingly, we observed that a significant number of genes (37%) that were reported to be down-regulated in the absence of H3.3 (H3.3-activated genes), are upregulated in the absence of ATAD2, particularly at day 26 PP, when post-meiotic cells are present in the testes (Fig. 5A [DOI](#), Fig. S1 [DOI](#)). Gene Set Enrichment Analysis (GSEA) applied to genes that are differentially expressed between wild-type and *Atad2* KO cells at day 26 PP, confirmed that the "H3.3-activated" gene set (corresponding to the published list of H3.3-activated genes) are clearly enriched in *Atad2* KO cells (Fig. 5A [DOI](#), right panel).

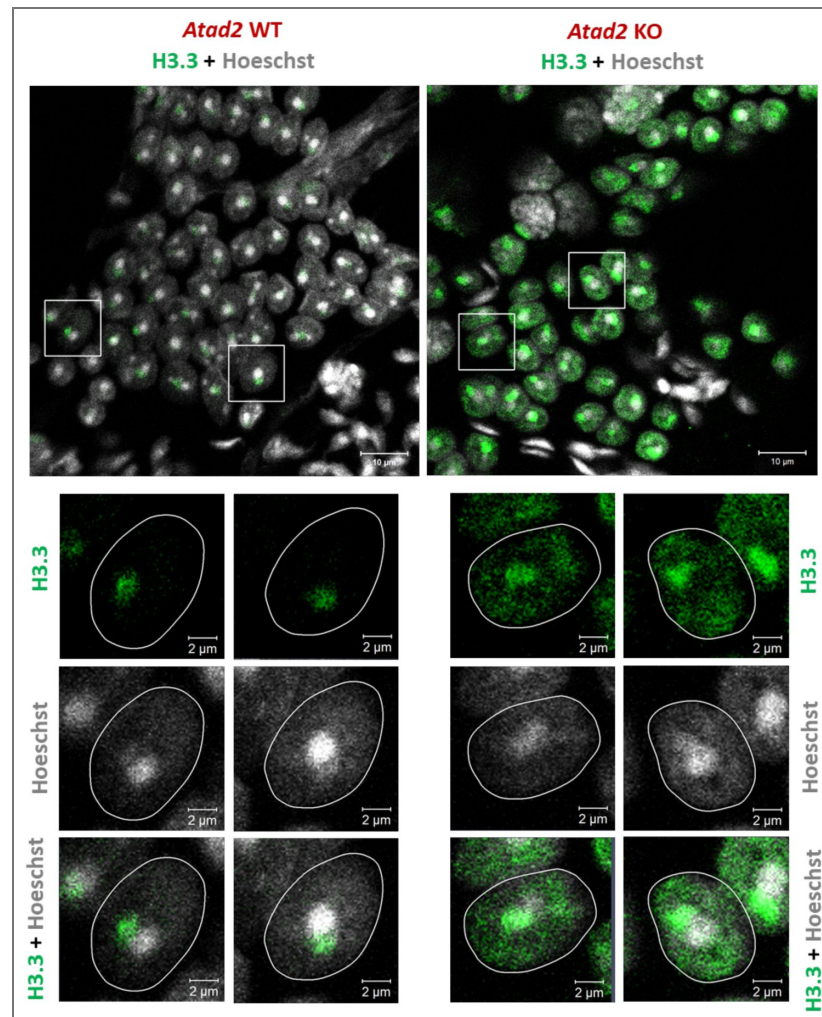


Figure 3. Enhanced H3.3 detection in *Atad2* KO post-meiotic cells.

A specific antibody against H3.3 was used to detect H3.3 in round spermatid cells from wild-type (*Atad2* WT) and knockout (*Atad2* KO) mice. Round spermatids are recognizable by their distinctive Hoechst - bright chromocenter (upper panels). The lower panels represent a higher magnification of selected cells that are displayed at lower magnification in the upper panels. The images show H3.3 labelling and Hoechst staining alone or the merged images of these staining as indicated. The nuclear boundaries defined by Hoechst staining, are outlined by white lines.

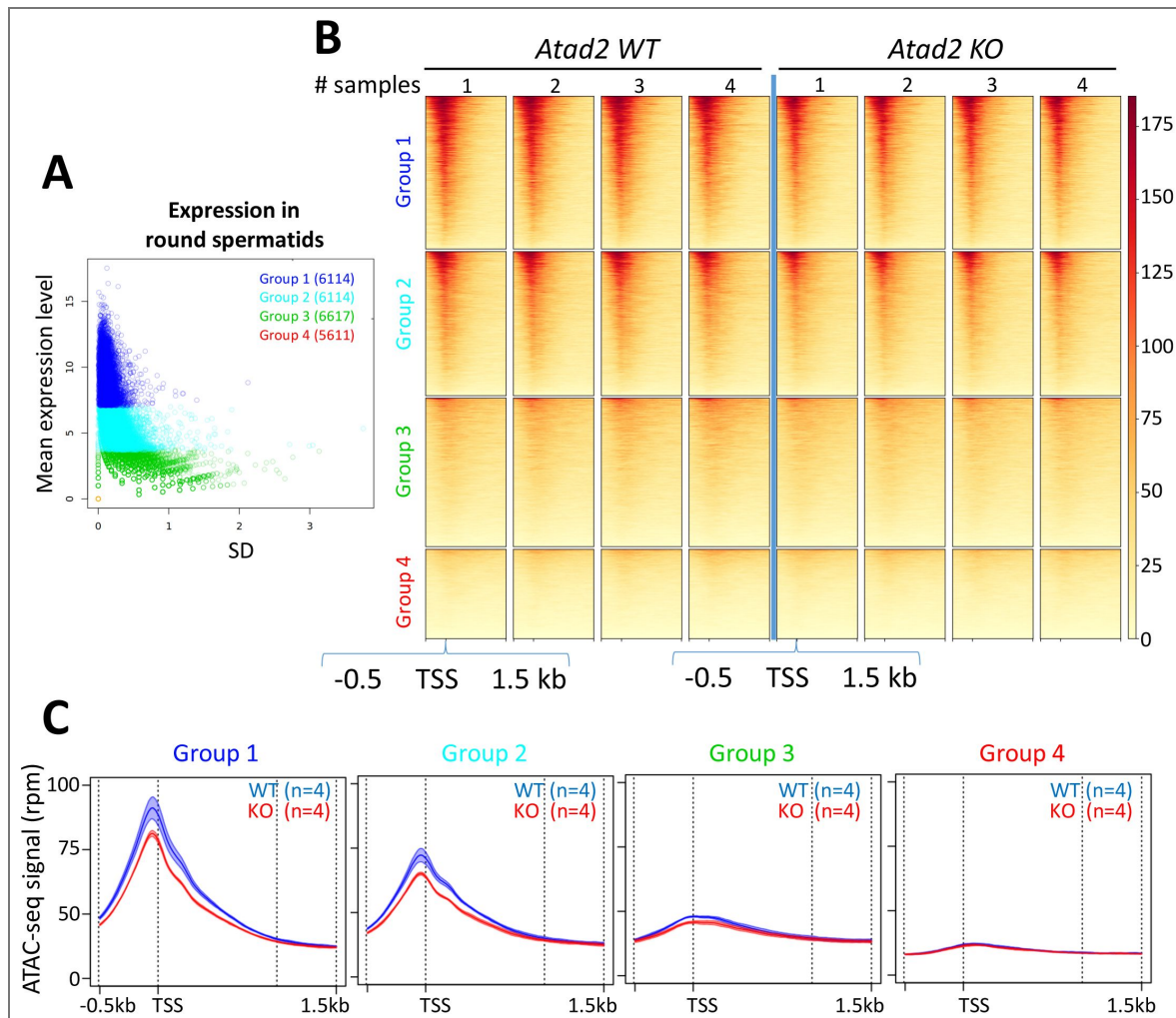


Fig. 4. ATAD2 controls active gene TSS accessibility.

A) Gene selection according to expression in wild-type round spermatids. The plot displays the mean expression levels (y-axis, labeled as “mean expression level”) versus the corresponding standard deviations (x-axis), both calculated from three independent biological replicates of isolated round spermatids. The standard deviation reflects the variability of gene expression across biological replicates. Genes were grouped into four categories (Group1: blue, Group 2: cyan, Group 3: green, Group 4: orange) according to the quartile of their mean expression. For Group 4, all genes have no detectable expression, resulting in a mean expression of zero and a standard deviation of zero; consequently, the 5611 genes in this group are represented by a single overlapping point (red open circle) at the origin. **B)** Heatmap on ATAC-seq signal. The heatmap shows the ATAC-seq signal of each group (Group 1 in blue, Group 2 in cyan, Group 3 in green and Group 4 in orange) in rows, for 4 *Atad2* WT and 4 *Atad2* KO samples in column. **C)** Profiles according to selected genes. The mean profiles are displayed together with their variability (± 2 s.e.m.) across the four replicates for both *Atad2* WT (blue) and *Atad2* KO (red). For groups 1, 2, and 3, the envelopes of the curves remain clearly separated around the peak, indicating a consistent difference in signal between the two conditions. In contrast, group 4 does not present a strong signal and, accordingly, no marked difference is observed between *Atad2* WT and KO in this group.

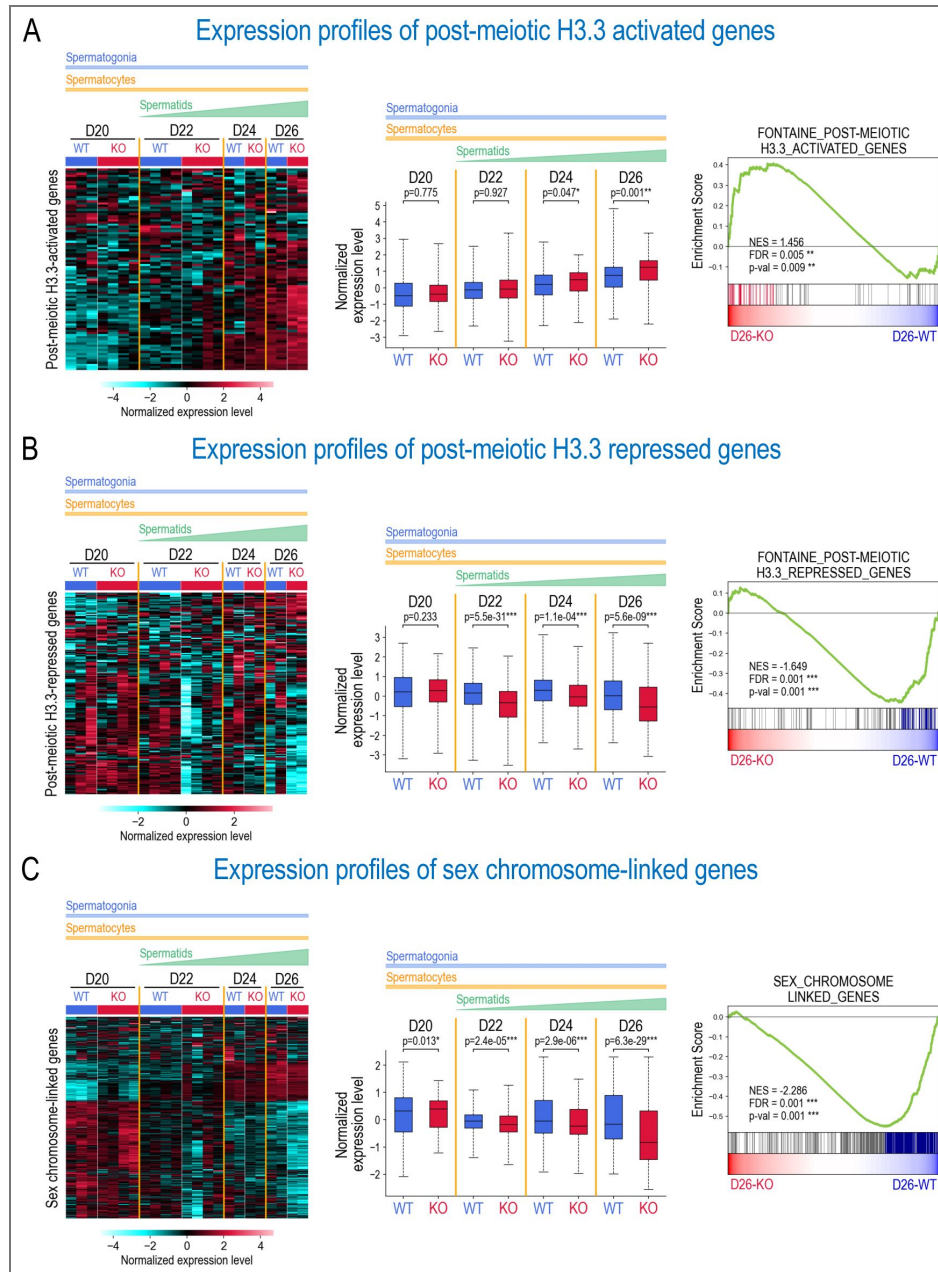


Figure 5. ATAD2 enhances H3.3 gene regulatory functions.

(A) Expression profiles of post-meiotic H3.3-activated genes. The heatmap (left panel) displays the normalized expression levels of genes identified by Fontaine and colleagues as upregulated in the absence of histone H3.3 (Fontaine et al. 2022) for *Atad2* WT (WT) and *Atad2* KO (KO) samples at days 20, 22, 24, and 26 PP (D20 to D26). The colour scale represents the z-score of log-transformed DESeq2-normalized counts. The middle panel box plots display, pooled, normalized expression levels, aggregated across replicates and genes, for each condition (WT and KO) and each time point (D20 to D26). Statistical significance between WT and KO conditions was determined using a two-sided t-test, with p-values indicated as follows: * for p-value<0.05, ** for p-value<0.01 and *** for p-value<0.001. The right panel shows the results of gene set enrichment analysis (GSEA), which assesses whether predefined groups of genes show statistically significant differences between conditions. Here, the post-meiotic H3.3-activated genes set, identified by Fontaine et al. (2022), is significantly enriched in *Atad2* KO compared with WT samples at day 26 ($p < 0.05$, $FDR < 0.25$). Coloured vertical bars indicate the “leading edge” genes (i.e., those contributing most to the enrichment signal), located before the point of maximum enrichment score. **(B)** As shown in (A) but for the “post-meiotic H3.3-repressed genes” gene set. **(C)** As shown in (A) but for the “sex chromosome-linked genes” gene set.

This regulation can also be observed when we considered the published list of genes that were up-regulated in the absence of H3.3 (H3.3-repressed genes). In this case, the absence of ATAD2 led to the enhancement of the down-regulation of these genes (37%). In other words, the absence of ATAD2 led to an enhancement of H3.3-dependent gene repression (Fig. 5B [↗](#), Fig. S1 [↗](#)).

Fontaine and colleagues also reported that sex chromosome-linked genes that normally undergo Meiotic Sex Chromosome Inactivation (MSCI), were up-regulated in the absence of H3.3, indicating that H3.3 is required for their repression. We also investigated these genes and showed that they present an enhancement of repression in the absence of ATAD2 (51%) (Fig. 5C [↗](#), Fig. S1 [↗](#)).

These analyses clearly indicate that the absence of ATAD2 enhances H3.3 functions in agreement with increased residence time of H3.3 in nucleosomes. Genes that require H3.3 to be repressed or activated present an enhanced repression or activation, respectively, in the absence of ATAD2. Overall, these investigations remarkably confirmed a role for ATAD2 in regulating H3.3 function.

ATAD2 is required for controlled histone eviction and PRM assembly

ATAD2's involvement in HIRA-chromatin interaction mobility and its role in H3.3 assembly in post-meiotic cells, along with its similar function in the control of other histone chaperones like FACT (Wang et al. 2021 [↗](#)), led us to investigate whether ATAD2 participates in controlling the histone-to-protamine replacement that occurs later in these cells. The histone eviction process involves a genome-wide histone H4 hyperacetylation (Shiota et al. 2018 [↗](#)) and a TP-mediated PRMs assembly, which directs histone eviction and the final packaging of the male genome (Barral et al. 2017a [↗](#)). To test the involvement of ATAD2 in spermatogenic cell chromatin reorganization, we sought to co-detect testis-specific H2B variant, TH2B (Montellier et al. 2013 [↗](#)), along with transition protein 1 (TP1) and protamine 1 (PRM1). TH2B is first expressed at the commitment of spermatogenic cells into meiotic divisions and progressively becomes the major H2B species of spermatogenic cell chromatin in the subsequent stages (Montellier et al. 2013 [↗](#)). As expected, the accumulation of TP1 and PRM1 in wild-type cells is associated with the eviction of the majority of histones, reflected by the mutually exclusive labelling of TH2B and TP1/PRM1 in distinct cells (Fig. 6A [↗](#) and 6B [↗](#), *Atad2* wild-type mice). In contrast, in *Atad2* KO cells, TH2B and TP1/PRM1 were co-detected in a significant number of cells (Fig. 6A [↗](#) and 6B [↗](#)), suggesting that the process of histone eviction is disturbed. However, in *Atad2* KO cells, the initial delay in histone removal is transient, as spermatids with high levels of transition protein TP1 and PRM1 eventually appear (Fig. 6 [↗](#)). These observations strongly suggest that, in the absence of ATAD2, the process of histone eviction is also slowed down or delayed, allowing for the simultaneous detection of histones and histone-replacing proteins in the same cells.

Given that NUT-CBP/p300-dependent H4 hyperacetylation, along with the expression of H2A.L.2, TPs, and PRMs, is essential for histone removal, we also investigated whether the observed delay in histone removal is due to defective histone acetylation, particularly at H4K5, as well as disrupted cell type-specific expression and localization of H2A.L.2, TPs, and PRMs.

Immunohistochemical detection of H4K5ac, H2A.L.2, TP2, PRM1, and PRM2 revealed a similar pattern and cell type-specific expression of the investigated proteins when comparing *Atad2* KO and wild-type post-meiotic cells. (Fig. S2 [↗](#)).

These data support the idea that the perturbed modified flexibility of histone/TP/PRM chaperones does not grossly affect the other molecular events involved in the histone-to-PRM replacement.

Perturbed pre-PRM2 processing in the absence of ATAD2

We have shown previously that the H2A.L.2-dependent process of incorporation of TPs into nucleosomes is required for the proper processing of pre-PRM2. The abnormal overlapping staining of histones, TP and protamines observed by immunofluorescence in *Atad2* KO cells, suggested that TPs may not effectively mediate the pre-PRM processing. To test this hypothesis, we took advantage of our specific pre-PRM2 antibody (Rezaei-Gazik et al. 2022 [↗](#)) to investigate pre-PRM2 maturation and assembly. Figure 7A [↗](#) shows that although pre-PRM2 is present in both

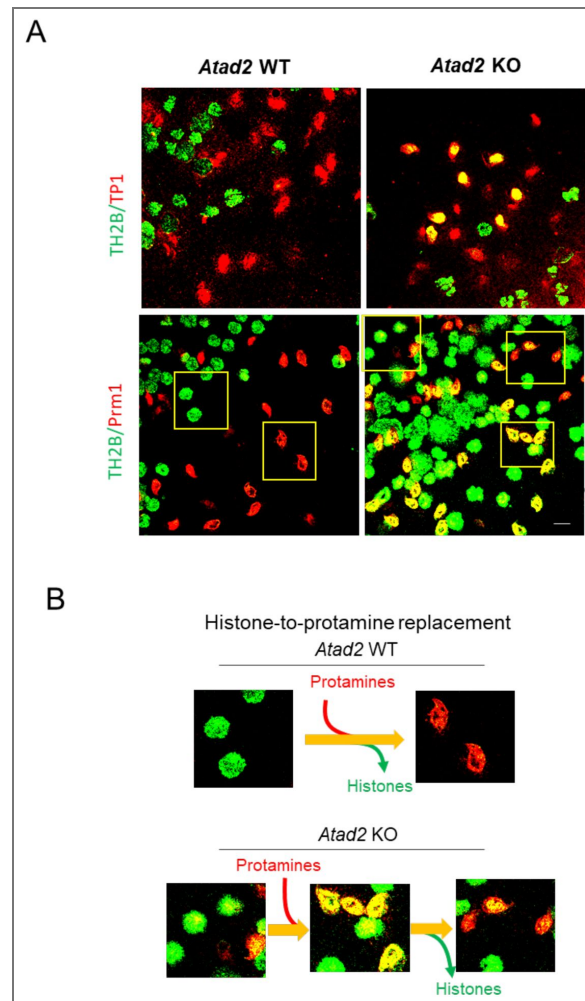


Figure 6. ATAD2 is required for efficient histone-to-protamine replacement.

(A) Spermatogenic cell preparations from wild-type and *Atad2* KO mice were stained with antibodies against TH2B (in green) and transition protein 1 (TP1) or protamine 1 (PRM1) (in red). A yellow signal indicates the co-detection of TH2B along with TP1 or PRM1. The indicated fields are shown at higher magnification in panel “B”. Scale bar: 10 μ m. (B) Fields shown in panel “A” are ordered to illustrate the transition from the histone associated genome to its packaging by protamines, considering both wild-type and *Atad2* KO spermatogenic cells.

wild-type and *Atad2* KO elongating spermatids, in the absence of ATAD2, pre-PRM2 tends to form dense aggregates as well as fragmented genomic domains. This phenomenon is also associated with an increased accumulation of pre-PRM2 protein level in testis extracts of *Atad2* KO spermatogenic cells compared to wild-type cells as revealed by ELISA and immunoblotting (Fig. 7B [↗](#), Fig. S3 [↗](#)). These data provide strong evidence that ATAD2 is required to streamline the activity of histone chaperones as well as other factors involved in histone exchange and TP/PRM assembly.

Defective mature spermatozoa genome organization and male fertility *in vitro*

We show that although the *Atad2* KO post-meiotic cells' defective chromatin organization does not affect the testis histology (Fig. S2 [↗](#)), it impairs testis weight and sperm count, which are significantly reduced in *Atad2* KO mice (Fig. S4 [↗](#)).

As expected, the impact of ATAD2 on HIRA function and TP/PRM assembly also has an impact on proper sperm genome compaction. Indeed, mature sperm genome decompaction test revealed a significant difference in the ability of the genome to resist genome decompaction conditions between wild-type *Atad2* and *Atad2* KO spermatozoa (Fig. 8A [↗](#); Fig. S5 [↗](#)). Given that spermatozoa with defective genome compaction are known to have poor success rate in *in vitro* fertilization (Bashiri et al. 2021 [↗](#)), we expected that spermatozoa from *Atad2* KO mice would perform worse in *in vitro* fertilization assays compared to wild-type spermatozoa. Figure 8B [↗](#) shows that this is, in fact, the case. However, surprisingly, fertility tests carried out on sexually mature adult males revealed that *Atad2* KO male mice are fertile, with only a statistically non-significant effect observed on their ability to foster pups (Fig. 8C [↗](#)), indicating that *in vivo* fertilization could buffer a relatively large series of sperm genome compaction defaults.

Discussion

Taking advantage of the very conserved structural features of ATAD2 and relying on the power of yeast genetics, we previously discovered that ATAD2's function is mostly to control major histone chaperone functions. Using a screen of the Abo1/ATAD2-dependent arrest of *S. pombe* cell growth, we found that impressively, 11 of the 14-growth arrest suppressor strains that we isolated, impacted histone chaperones or histones themselves, with 8 distinct inactivating mutations in the histone chaperone HIRA complex. Furthermore, investigation of ATAD2-depleted mouse ES cells showed that HIRA accumulates and remains bound to nucleosomes, leading to nucleosome overloading around active gene TSSs and on the normally nucleosome-free regions (NFRs) of these genes.

Here, we showed that *Atad2* is highly expressed in post-meiotic spermatogenic cells and that in its absence, as in ES cells and in different cancer cells, HIRA accumulates. Our new data suggest that in the absence of ATAD2 activity, HIRA residency time on chromatin increases leading to decreased turnover and overloading of its substrate, the histone variant H3.3 (Fig. 9 [↗](#)). This replication-independent histone has been first shown to assemble on the transcriptionally inactive meiotic cell sex chromosomes by Peter de Boer's laboratory (van der Heijden et al. 2007 [↗](#)). Subsequent studies showed that H3.3 remained enriched on the sex chromosomes also in early post-meiotic cells, usually localized close to the unique large chromocenter in these cells (Fontaine et al. 2022 [↗](#)).

In agreement with the proposed role for ATAD2 in the control of HIRA-dependent dynamics of H3.3, we observed a more intense labelling of H3.3 on the sex bodies in round spermatids and impressively, an increase in the genomic background detection of H3.3 in these cells.

The role of ATAD2 in the HIRA/H3.3 functions is further supported by comparing the reported H3.3-dependent transcriptome with the ATAD2-dependent transcriptome identified here.

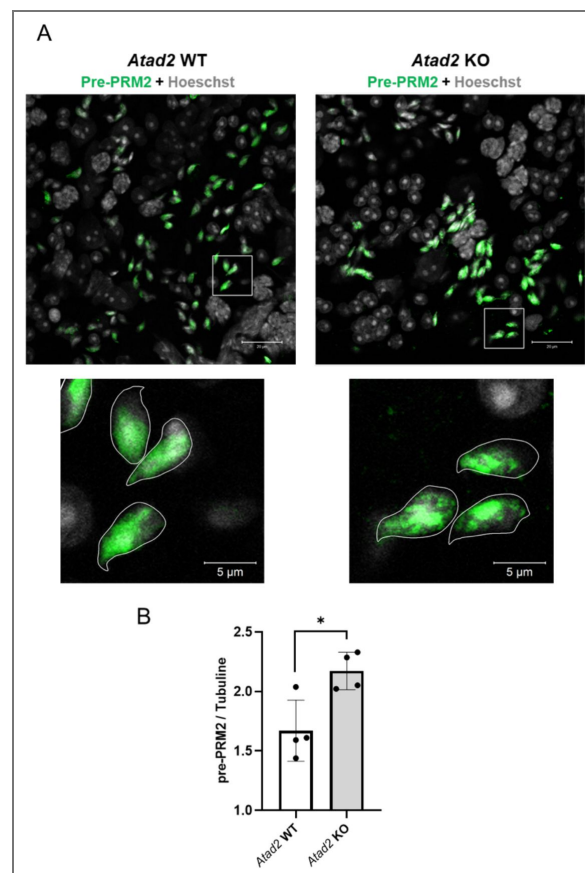


Figure 7. ATAD2 is required for proper pre-protamine 2 processing and protamine assembly

(A) A specific antibody raised against the processed part of pre-PRM2 (Rezaei-Gazik et al. 2022) was used to detect pre-PRM2 expression and localization in tubular elongating spermatid cells. The lower panels represent the magnification of selected cells. **(B)** The amount of pre-PRM2 was detected using soluble extracts from testis of wild-type and *Atad2* KO mice using ELISA. Histograms show the normalized values (pre-PRM2/tubulin) of 4 measurements of independent sedimentation samples of Elongating/Condensing fractions. Mean ± Standard Deviation is 1.6 ± 0.2 and 2.1 ± 0.1 for wild type and *Atad2* KO samples, respectively. The p-value obtained using an unpaired Student's *t*-test was 0.0158.

Figure 8. ATAD2s' function in sperm genome compaction and fertility parameters

(A) Epididymal sperm cells were isolated by the swim-up method from wild-type or *Atad2* KO mice and underwent a decompaction test and the sizes of the sperm heads were measured and represented as box plots ($n = 75$ sperm heads / box plot). The value ranges (mean) were 10.4 ~ 19.3 (16.0) for wild type sperms without decompaction; 11.4 ~ 20.1 (16.4) for *Atad2* KO sperms without decompaction; 33.4 ~ 61.8 (48.5) for wild type sperms after decompaction; 45.0 ~ 78.9 (62.6) *Atad2* KO sperms after decompaction. Tukey's HSD post two-way ANOVA test was used. **** $p < 0.001$. (B) Pools of oocytes from C57BL6 females were obtained and used for *in vitro* fertilization (IVF) with spermatozoa from WT or *Atad2* KO males (the experiment was repeated 4 times independently). The fertilization success rate is indicated as a percentage of oocytes giving rise to stage 2 embryos. Mean $\pm$ Standard Deviation is 25.5 ± 14 and 12 ± 5.2 for wild type and *Atad2* KO experiments respectively and the $p = 0.002$ for ratio paired Student t-test. (C) Five wild-type and five *Atad2* KO male mice were each mated with two C57BL6 females, the resulting litter sizes were recorded and represented as box plots ($n = 28$ and 30 , respectively). The average numbers of pups are respectively 8.0 ± 2.1 and 7.1 ± 2.5 and the p -value of Student t-test = 0.1 .

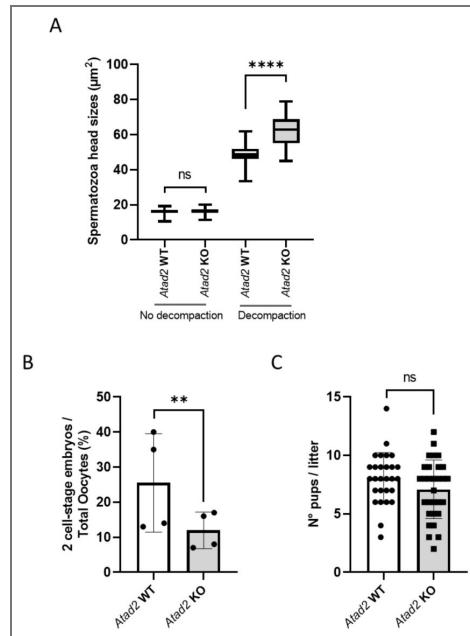
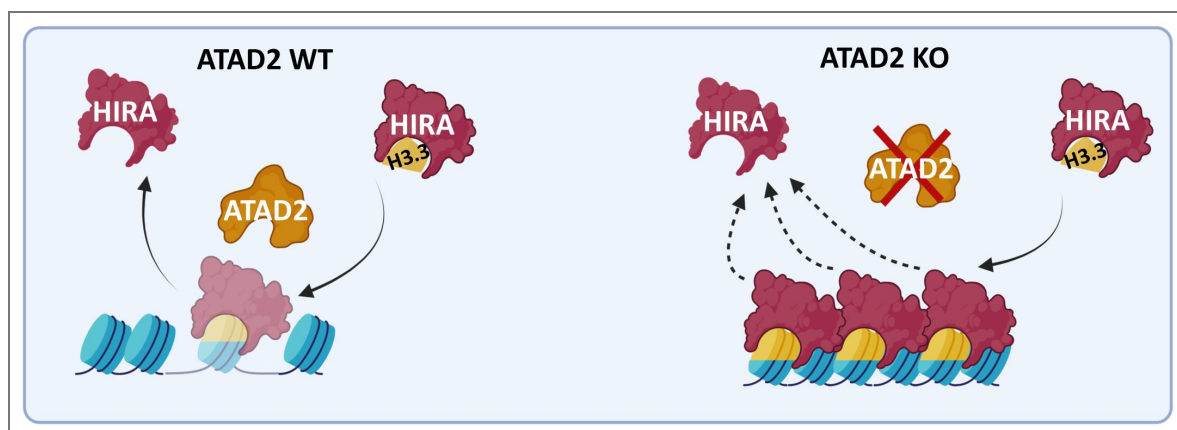


Figure 9. A model summarizing the role of ATAD2 in regulating chromatin-bound HIRA dynamics (along with other histone and non-histone protein chaperones) during the post-meiotic stages of mouse spermatogenesis.

Following the deposition of H3.3 by HIRA onto chromatin, ATAD2 facilitates the release of HIRA. In the absence of ATAD2, both HIRA and the newly incorporated H3.3 persist longer in nucleosomes. The scheme was created using BioRender.com/cmcjckk.



Indeed, in the absence of ATAD2, we observed an enhancement of H3.3-dependent gene regulatory events (activation or repression), in agreement with an increased H3.3 presence and hence function, in the chromatin of *Atad2* KO post-meiotic cells.

These investigations on ATAD2's role in post-meiotic spermatogenic cells built upon our previous findings in *S. pombe* and ES cells, uncover an additional level of regulation of histone turnover, which relies on the modulation of histone chaperone–chromatin interactions. Previously, A. Groth's laboratory discovered that the histone chaperone DNAJC9 recruits the ATPase HSP70 to provide the necessary energy to release the chaperone from chromatin (Hammond et al. 2021 [↗](#)). Considering our data on the role of the ATPase ATAD2 with respect to HIRA and that of HSP70 with respect to DNAJC9, it is tempting to propose that ATP energy is required for the proper function of histone chaperones in controlling histone turnover.

Interestingly, very recently, Song laboratory demonstrated that the *S. pombe* Abo1/ATAD2, is required for ATP-dependent dissociation of the histone chaperone FACT from chromatin, particularly at TSSs (Jang et al. 2024 [↗](#)). In line with this conclusion, ChIP-seq mapping of FACT in ATAD2-depleted ES cells revealed an accumulation of nucleosome-bound FACT at the TSSs of active genes (Wang et al. 2021 [↗](#)), supporting the idea that ATAD2 not only regulates the dynamics of histone-bound HIRA but also that of other histone chaperones such as FACT.

In addition to histones, many chromatin-bound regulatory factors also need to dynamically interact with chromatin and hence ATAD2 could also be involved in mediating the dynamic interactions of such chromatin-bound factors. In agreement with this hypothesis, using proteomics of ATAD2-bound nucleosomes in ES cells, we previously co-purified various chromatin bound factors (Morozumi et al. 2016 [↗](#)). Other AAA ATPase factors, such as p97/VCP, have also been implicated in the remodeling of chromatin-bound factors. Indeed, the AAA ATPase factor, p97/VCP, has been shown to remove H3 variant CENP-A (van den Berg et al. 2023 [↗](#)), the chromatin trapped PARP1 (Krastev et al. 2022 [↗](#)), trapped Ku70/80 (van den Boom et al. 2016 [↗](#)), DNA damage sensors, DDB2 and XPC (Puumalainen et al. 2014 [↗](#)), RNA pol II (Lafon et al. 2015 [↗](#)), and other chromatin interacting factors. Similarly, ATAD2 AAA ATPase very likely acts on other chromatin-bound factors than histone chaperones. However, in contrast to p97/VCP, and due to its bromodomain, ATAD2 would preferentially act on H4K5 acetylated chromatin regions and would specifically ensure the dynamics of H4 acetylated chromatin-bound factors. Based on the data presented here, we can also include, a role for ATAD2 to streamline steps involving the H2A.L.2-dependent recruitment of TPs and protamine assembly and the final histone displacement in spermatids. The chaperones responsible for TP/PRM assembly are not yet identified, but it is possible that ATAD2 plays a role in dissociating these chaperones from their substrate proteins after the assembly process. That is why in our *Atad2* KO spermatids, we observed an overlap between TP/PRM and histones. This could be due to the delayed removal of chaperones bound to TPs and PRMs and the inability of pre-PRM2 to be processed and hence of PRMs to efficiently replace histones. Since histone replacement still occurs in the absence of ATAD2, it suggests that other AAA ATPases, such as p97/VCP, may compensate for its function. However, under these conditions, although most of the histone replacement takes place, our functional tests such as *in vitro* chromatin decompaction and *in vitro* fertilization, strongly suggest that mature spermatozoa genome organization is not fully complete and the spermatozoa nuclei remain fragile and sensitive to external assaults.

In summary, the functional analysis of ATAD2 during its physiological expression in the post-meiotic phases of spermatogenesis supports our previous conclusions regarding its role in ES cells and its conserved activity in *S. pombe*. Together, these studies reveal a novel layer of regulation in histone turnover and chromatin dynamics, based on the control of histone-bound histone chaperone release.

Materials and Methods

Animal care and breeding

Mice were housed at the Grenoble High Technology Animal Facility (PHTA). Mice were euthanized following a procedure approved by the official ethic committee of the University Grenoble Alpes (COMETH, C2EA-12). The investigators directly involved in care and breeding of mice had an official animal-handling authorization obtained after 2 weeks of intensive training and a final formal evaluation.

Generation of *Atad2*-KO mice

Atad2 KO mice were obtained from the International Knockout Mouse Consortium, the Wellcome Trust Sanger Institute, UK. The 'knockout-first' allele (tm1a) contains an IRES: *lacZ* trapping cassette and a floxed human β -actin promoter-driven *neo* cassette inserted into the intron 12 of the *Atad2* gene (Skarnes et al. 2011 [DOI](#)). We created *Atad2*^{11 Neo-Exon 12} mice by crossing mice bearing the tm1a allele with mice expressing Cre recombinase under the control of a ubiquitous human promoter CMV. This cross results in an *Atad2*^{LacZ11Neo-Exon12} knockout mouse with the first 11 exons of *Atad2* being fused to β -gal, leading to a non-functional ATAD2 protein depleted of its bromodomain and AAA ATPase domain.

For *Atad2* KO mice genotyping, multiplex PCR was used to detect LacZ cassette producing a 393-bp WT band and a 293-bp mutant band using primers *Atad2*_84231_F, TATCCAACAAGCCTGAGCCC, *Atad2*_84231_R, CAACTGGAGCTGGGTCTTCC and Cas-R, TCGTGGTATCGTTATGCGCC.

Fertility test

Fertility tests were carried out with sexually mature 3-month-old males. Five WT males and five *Atad2* KO males were crossed with two C57BL6 females each, during 3 months, and the litter sizes were monitored during this period.

In vitro Fertilization

Eggs were obtained from 6- to 7-week-old superovulated C57/BL6J females, stimulated by a first injection of 7.5 UI of PMSG (Pregnant Mare Serum Gonadotropin), followed 48 hours later by a second injection of 7.5 UI of HCG (Human Chorionic Gonadotropin). Fourteen hours later, females were euthanized by cervical dislocation and cumulus-oocytes complexes (COCs) released from the ampulla were collected in 500 μ L M2 medium. Spermatozoa from wild type and *Atad2* KO males were harvested by dilaceration of the cauda epididymitis and allowed to swim in 1 mL M2 medium for 10 min at 37°C. Two hundred microliters of sperm were transferred into 800 μ L M16 /2% BSA and capacitated for 80 min at 37°C /5% CO₂. Finally, 300,000 sperm were simultaneously added to the COCs and incubated in M16 medium at 37°C/5% CO₂. After 4 hours of incubation, unbound spermatozoa were removed by 3 successive washes with 500 μ L of M16. Twenty-four hours after fertilization, unfertilized eggs and two-cell embryos (as an indication of successful fertilization), were scored.

Protein extraction and Immunoblotting

Proteins were extracted in 8M Urea, sonicated at 150J with a probe sonicator, mixed with 4x Laemli loading buffer and heated for 5 minutes at 95 °C. Protein samples were loaded on 4-12% Bis-Tris gels and migrated at 150 V for 1h30. The transfer step was realized using a nitrocellulose membrane with the Bio-Rad Transfer Turbo machine. Nitrocellulose membrane was blocked in 5% milk PBS-Tween 0.1% solution, washed 3 times for 10 minutes with PBS-Tween 0.1% solution and incubated overnight at 4°C with specific antibodies listed in [Table S1](#) [DOI](#). Membranes were washed 3 times for 10 minutes with PBS-Tween 0.1% solution and were incubated with HRP secondary antibodies at 1/10000 dilution for 1h at RT under agitation. Revelation was performed using ECL substrates and visualization was performed with the Vilber Fusion FX Chemiluminescence imaging system (Vilber).

ELISA

Urea protein extracts were diluted in PBS and used to coat 96-well round-bottom plates, followed by overnight incubation at 4 °C. Coating solution was removed and wells were washed 3 times with washing buffer (BSA 1%, PBS-T 0,1%). Wells were blocked with blocking buffer (5% BSA in PBS-T 0,1%) for 2 hours at room temperature under agitation. Blocking buffer was removed and wells were washed 2 times with washing buffer. Primary antibodies for pre-PRM2 (1:1000) and tubulin (1:5000) were diluted into blocking buffer, pipetted into their corresponding wells and incubated overnight at 4°C under mild agitation. Wells were washed 4 times with washing buffer and subsequent incubation with secondary HRP antibodies, diluted 1:5000 in blocking buffer, was performed for 1 hour at room temperature under mild agitation. Wells were washed 4 times with washing buffer, and TMB ELISA Substrate was added to wells and incubated until the color developed, at which time Stop Solution was added to the wells to stop the reaction. The signal was quantified using a plate reader at 450nm.

X-Gal staining

Male mice were euthanized by cervical dislocation. Testes were dissected out, drilled in 3 points and incubated overnight at 4°C under agitation in fixative solution (0.5% glutaraldehyde, 2 mM MgCl₂ and 5 mM EGTA in PBS). After incubation until equilibration in an equilibration buffer (30% sucrose, 2 mM MgCl₂ and 5mM EGTA in PBS), testes were frozen individually in OCT on dry ice. Frozen blocks were sectioned at 10 µm slices on Superfrost™ slides and let dry overnight. Slides were fixed with fixation buffer (0.2% glutaraldehyde, 100 mM MgCl₂ and 5 mM EGTA in PBS) for 10 min, washed with washing buffer (PBS, 0.02% NP-40, 0.01% Sodium deoxycholate and 2 mM MgCl₂) for 5min and incubated three times in 50% Ethanol for 5 min each. Then the slides were stained with staining solution (2 mM MgCl₂, 0.02% NP-40, 0.01%Na Doc, 5 mM potassium ferricyanide, 10 mM potassium ferrocyanide and 0.5 mg/ml X-gal) in a wet chamber for several hours at 37°C. The slides were then washed with washing buffer for 5 min and with distilled water, and a counterstaining was performed with hematoxylin. Finally, the slides were washed with running tap water for several minutes and mounted with Dako fluorescent mounting medium.

Immunofluorescence staining of germ cells

Staged seminiferous tubules were prepared as detailed in [Gaucher et al. 2012](#) [\[1\]](#). In order to permeabilize the cells, the slides were placed in 0.5% saponin, 0.2% Triton X-100, and 1 × PBS at RT for 15 min and washed for 5 min with PBS. The preparations were then incubated in 5% milk, 0.2% PBS-Tween blocking buffer, at RT for 30 min under agitation. The primary antibodies were diluted in 1% dry milk, 0.2% PBS-Tween buffer. The slides were incubated overnight with this diluted solution of the primary antibodies in a humidified chamber at 4°C and then washed three times for 5 min each in the antibodies' dilution buffer. The secondary antibodies were diluted at 1:500 in the same buffer and incubated in a humidified chamber for 30 min at 37°C, and then washed as for the primary antibodies. The DNA was counterstained with Hoechst, and the slides were mounted using Dako fluorescent mounting medium.

Sperm head decompaction test

Spermatozoa from wild type and *Atad2* KO males were harvested by dilaceration of the cauda epididymis and allowed to swim in 1 mL M2 medium for 10 min at 37°C. The sperm-containing suspension was transferred into a 1.5 ml Eppendorf tube and centrifuged at 3000 rpm for 8 min in 4°C. The sperm pellets were resuspended in PBS and dropped onto glass slides to spread and dry at RT. The decompaction mix (50 mM DTT, 400 IU/ml heparin and 0.2% Triton in PBS) was added to each dried sperm drop for 2 min and the decompaction was stopped by placing the slides in 4% PFA for 15 min. Slides were washed with PBS for 5 minutes, stained with DAPI, and mounted using Dako fluorescent mounting medium. Fluorescent images were captured by a Zeiss inverted microscope under a 63X numeric aperture oil-immersion lens (Carl Zeiss). The size of spermatozoa heads was calculated with ImageJ software.

Spermatogenic cell fraction purification

Spermatogenic cell fraction purification was performed as previously described (Shiota et al. 2018 [↗](#)).

ATAC-seq and data analysis Library preparation

Frozen nuclei from four biologically independent purified spermatid fractions were centrifuged at 3000 g for 5 minutes and then washed in a 1% BSA-PBS solution. Libraries were then prepared from 100,000 nuclei using the ATAC-seq Kit (Catalog No. 53150) from Active Motif, following the manufacturer's protocol (version B9). After quality control (QC) with Qubit and Fragment Analyzer, libraries were subsequently purified using AMPure XP (Beckman Coulter, Catalog No. A63881) beads with a double size selection of 100-800 bp fragments. After a second round of library QC (Qubit and Fragment Analyzer) we performed a paired-end sequencing on a NextSeq 2000 device using a 200-cycle P2 flow cell.

Trimming

The raw fastq files were processed by 5' end trimming, keeping 30bp-length fragments, using fastx_trimmer [http://hannonlab.cshl.edu/fastx_toolkit/ [↗](#). Accessed 28 Feb. 2022.], with options -l 30 -Q33.

Alignment

The trimmed fastq files were aligned on the UCSC Mus_musculus mm10 genome using the Bowtie2 aligner (Langmead and Salzberg 2012 [↗](#)), with options -end-to-end, -no-mixed, -no-discordant.

Normalization

The BigWig files (.bw) containing normalized integrated aligned read count signals were obtained from by normalizing and smoothing bam files using bamCoverage (from deepTools suite (Ramirez et al. 2014 [↗](#)) with options: -binSize 4 -minMappingQuality 30 -normalizeUsing RPM.

Heatmap

The normalized ChIP signal were converted into a 10bp bin matrix of the signal 500b upstream and 1500b downstream protein-coding genes' TSS, using computeMatrix (from deepTools suite (Ramirez et al. 2014 [↗](#)), with options reference-point -R mm10_pc.bed -referencePoint TSS -binSize 10 -beforeRegionStartLength 500 -afterRegionStartLength 1500 Heatmaps were generated using plotHeatmap (deepTools suite (Ramirez et al. 2014 [↗](#)).

RNA-seq data processing

Wild-type (WT) and *Atad2* knockout (KO) homozygous males were sacrificed by cervical dislocation on days 20-, 22-, 24-, and 26-postpartum (PP), after which their testes were harvested. RNA was extracted using TRIzol™ reagent, followed by DNA digestion with DNase I. The resulting RNA samples were used for RNA sequencing. For both WT and *Atad2* KO mice, the biological replicates included 3 samples for D20, 4 samples for D22, 2 samples for D24 and 2 samples for D26.

The raw sequencing data (FASTQ files) were aligned to the UCSC mm10 mouse genome using the STAR software (version 2.7.11b) (Dobin et al. 2013 [↗](#)) to generate BAM files. Raw read counts were extracted from the BAM files using HTSeq (version 2.0.5) (Anders et al. 2015 [↗](#)) with the following parameters: -t exon, -f bam, -r pos, --stranded=reverse, -m intersection-strict, and --nonunique none.

Normalization of read counts was performed using the DESeq2 R package (version 1.22.2) (Anders and Huber 2010 [↗](#); Love et al. 2014 [↗](#)). Counts were transformed into DESeq2-normalized values and subsequently log-transformed using the formula $\log_2(1 + \text{DESeq2-normalized counts})$. Differential expression analysis was conducted with the R package "SARTools" (Varet et al. 2016 [↗](#)).

Purified RNA from fractionated round spermatids were also sequenced to establish gene transcriptional group quartile shown in Fig. 4. RNA purification and sequencing were performed as described above.

Quantification and statistical analysis

Statistical analysis was performed with R and GraphPad Prism 8. The results of fertility tests were analyzed by Student t-test. Pairwise Student t-test was used to analyze testis weights and spermatozoa counts of same-aged mice and ratio pairwise Student t-test was used for the IVF efficiency results. The results of sperm decompaction test were analyzed by two-way ANOVA and Tukey's HSD post two-way ANOVA test was used to determine the significance between groups. *P*-values < 0.05 were considered statistically significant: **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

Gene Set Enrichment Analysis (GSEA)

Differential expression analysis and the corresponding GSEA analysis (Mootha et al. 2003; Subramanian et al. 2005) were performed to compare *Atad2* KO versus *Atad2* wild-type samples at days 20, 22, 24, and 26 PP. For the GSEA analysis, we used the Python package “gseapy” (Fang et al. 2023) available at <https://www.gsea-msigdb.org/gsea>.

The GSEA was carried out on three custom gene sets. The first gene set, referred to as “sex chromosome-linked genes”, corresponds to genes located on the X and Y chromosomes. The remaining two gene sets include genes identified as upregulated or downregulated in the absence of histone H3.3 by Fontaine and colleagues (Fontaine et al. 2022). These gene sets were designated as “post-meiotic H3.3-activated genes” and “post-meiotic H3.3-repressed genes”, respectively.

Data availability

1) The transcriptomic data generated in this study, corresponding to testes collected at different days post-partum, have been deposited on GEO under the accession number GSE277943. 2) Transcriptomic data from fractionated round spermatids (three biological replicates, Fig. 4A) were deposited on GEO accession GSE284749. 3) ATAC-seq data generated in this study were deposited on GEO under the accession number GSE283879.

Supplementary materials

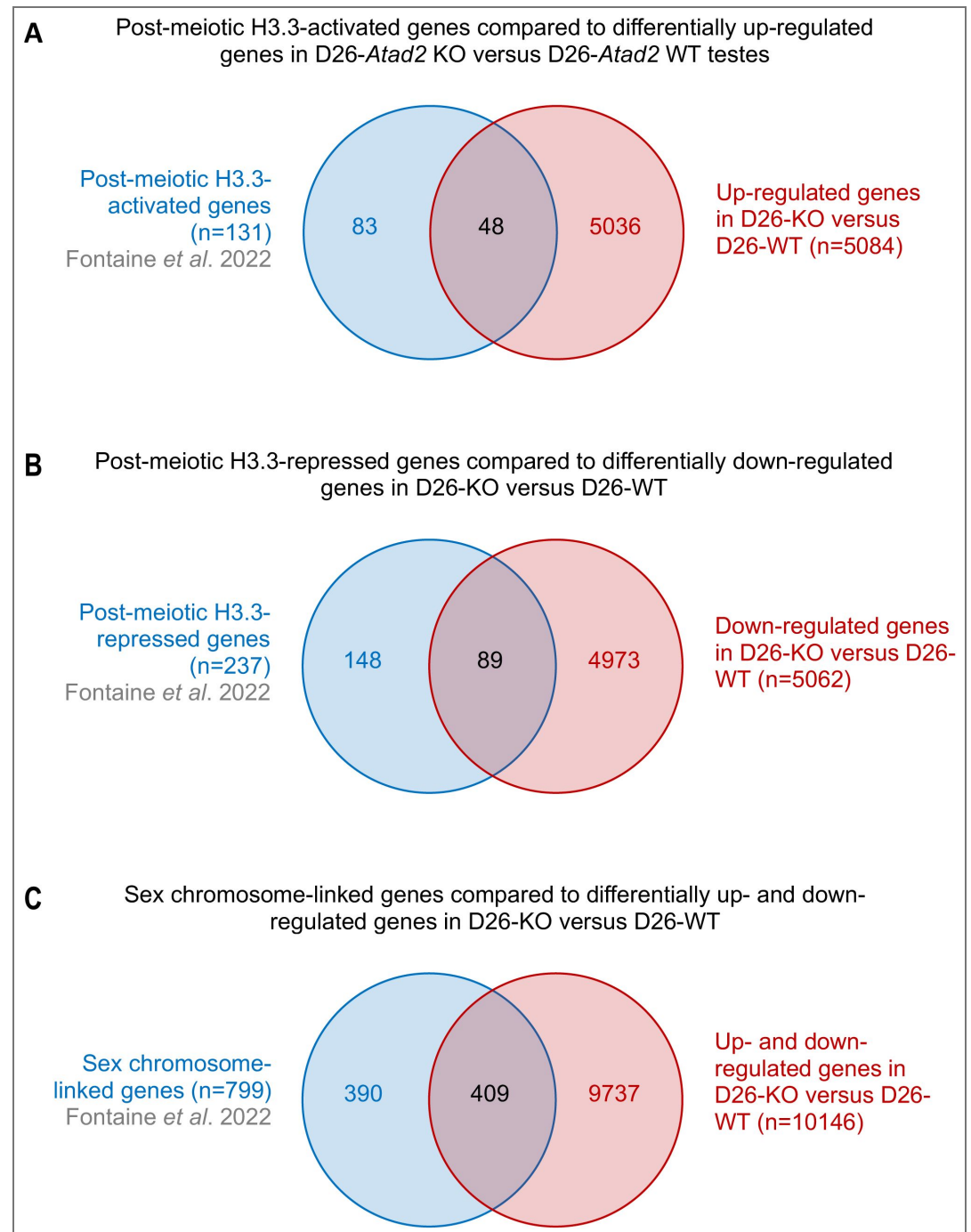


Figure S1. Venn diagrams showing the overlap of significantly differentially expressed genes between *Atad2* KO and *Atad2* WT samples at day 26 post-partum (PP), compared with three gene sets reported by (Fontaine *et al.* 2022). (A) Genes upregulated in the absence of histone H3.3 in both *Atad2* WT and *Atad2* KO samples at day 26 PP. (B) genes repressed under the same conditions, and (C) sex chromosome-linked genes. In the differential expression analysis, a fold-change greater than 1.2 between *Atad2* KO and *Atad2* WT samples was considered significant.

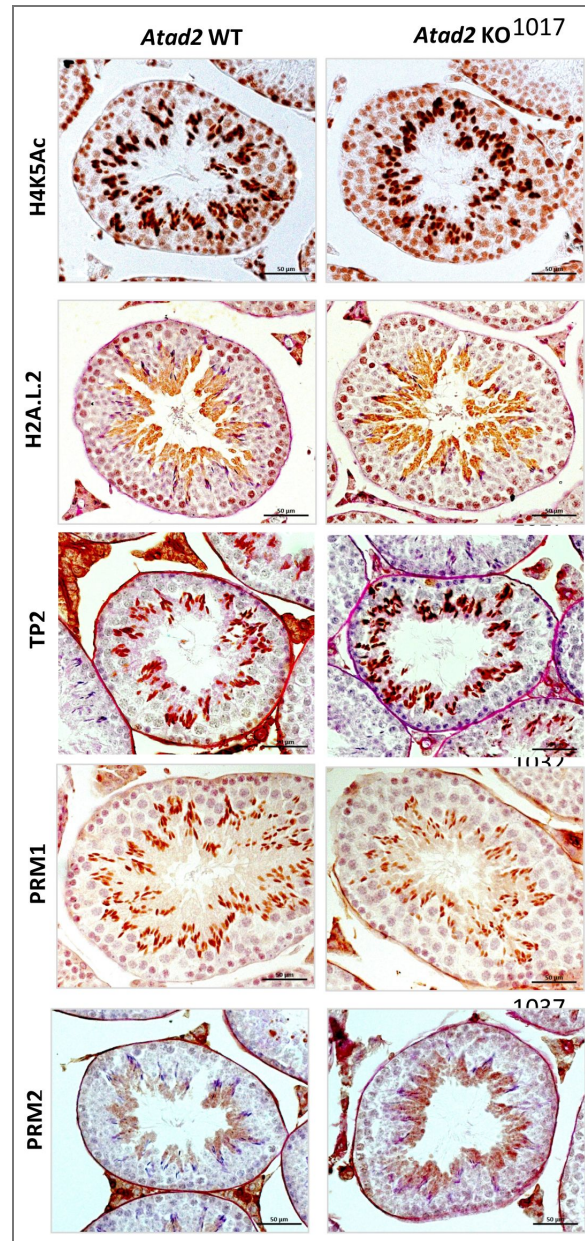


Figure S2. *Atad2* gene inactivation does not disrupt testis morphology, acetylation of H4K5ac, or the cell-specific accumulation of H2A.L.2, TP2, PRM1 and PRM2.

Paraffin-embedded sections from wild-type or *Atad2* KO testes were used to visualize the accumulation of H4K5ac or expression of H2A.L.2, TP2, PRM1 and PRM2.

Figure S3. ATAD2 is required for proper pre-protamine 2 processing.

Extracts from representative samples shown in Fig. 7B, derived from wild-type or *Atad2* KO cells, were analyzed by SDS-PAGE followed by immunoblotting using our anti-prePRM2 and anti-histone H3 antibodies. In each panel, the films were aligned with the corresponding blot and photographed. M, molecular weight marker.

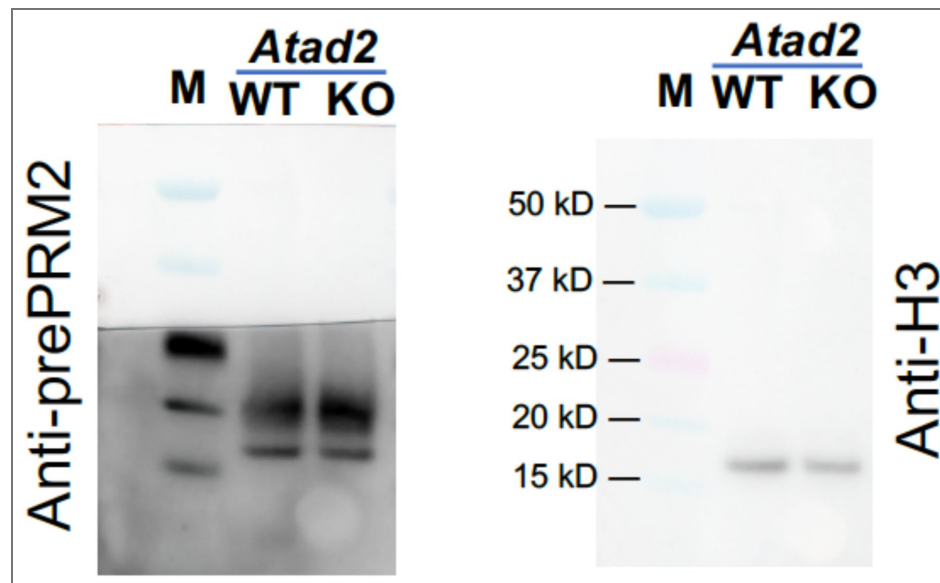
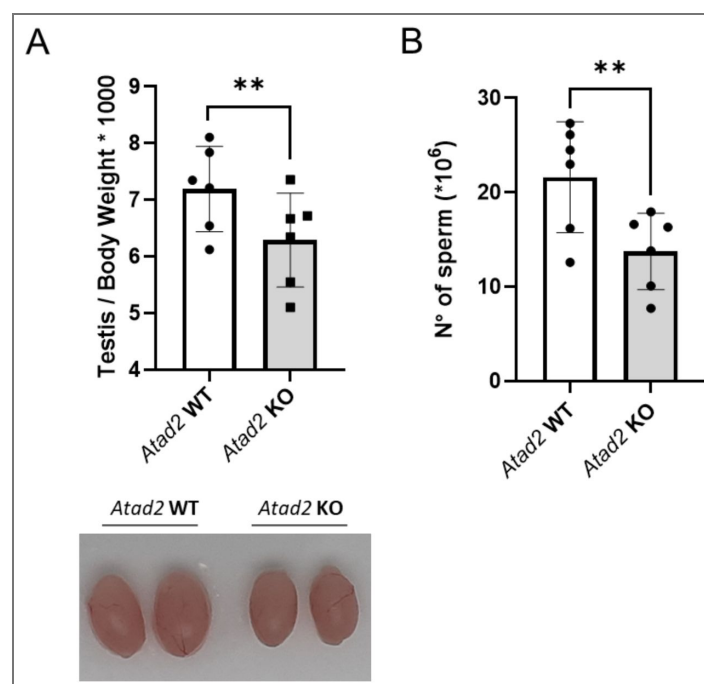


Figure S4. *Atad2* depletion affects testis weight and mature sperm counts.

(A) Testes from six 21- to 27-week-old wild-type and *Atad2* KO mice were collected, weighted (mg) and their weights normalized to body weight and only same-age mice were compared pairwise. Histograms represent the mean of the measured weights and error bars the Standard Deviation (upper panel). The *p*-value of pairwise Student t-test = 0.004. The lower panel shows representative testes from wild-type and *Atad2* KO mice. (B) Spermatozoa count (Cells /ml) were performed for pairs of wild type and *Atad2* KO (*n*=6). The average numbers of spermatozoa counts are respectively 21.6 ± 5.8 and $13.7 \pm 4.$, and the *p*-value of paired Student t-test = 0.001.



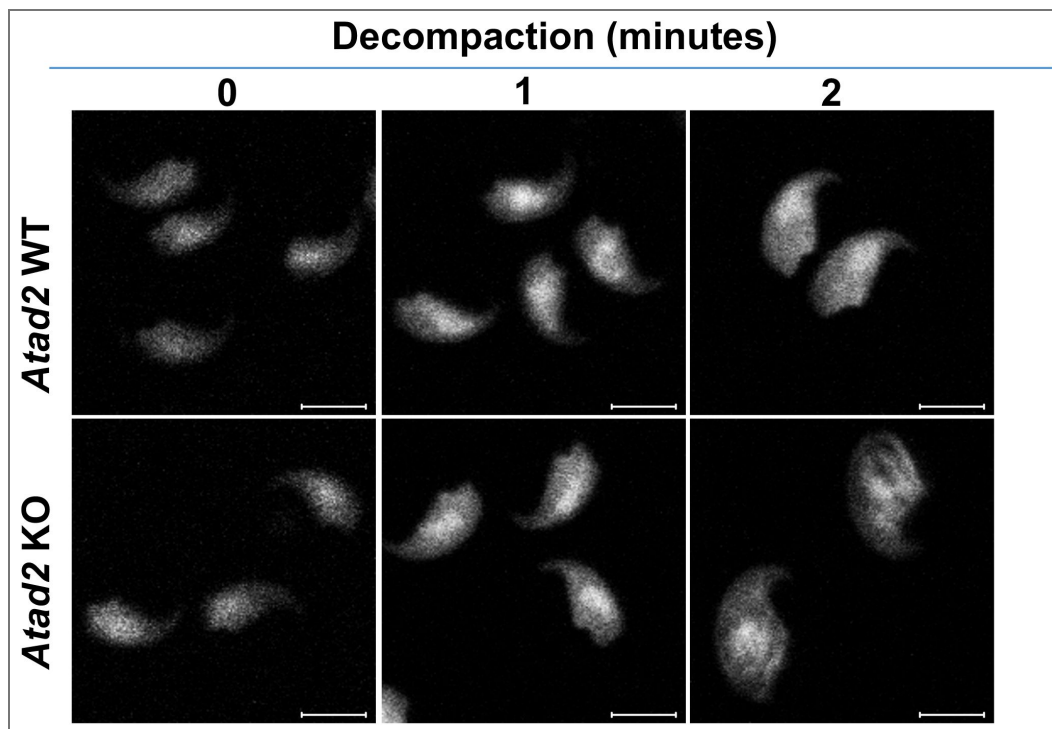


Figure S5. Sperm head decompaction after DTT and heparin treatment in wild-type and *Atad2* KO males.

Epididymal sperm were treated with a DTT/heparin/Triton mixture, fixed, and stained with DAPI. Fluorescent images were acquired using a 63× oil-immersion objective, and sperm head size was quantified with ImageJ. Scale bars represent 5 μ m.

Name	Company	Reference	Host Species	Dilution
Anti-ATAD2	Homemade	-	Rabbit	1:1000
anti-H2A.L1/L2 SAB771	Homemade	-	Rabbit	1:200 for IHC
Anti-Histone H3 antibody - Nuclear Marker and ChIP Grade	Abcam	Ab1791	Rabbit	1:5000 for WB
Anti-H2B (testis specific)	Abcam	AB178426	Rabbit	1:100 for IF
anti-H3F3B	Abnova	#H00003021-M01	Mouse	1:200 for IF
Anti-H4K5Ac	PTM Biolabs	#PTM-163	Mouse	1:200 for IHC
Anti-HIRA	Cell Signaling Technology	D6O8L	Rabbit	1:1000 for WB
Pre-PRM2	Homemade	-	Rabbit	1:100 for IF
anti-Protamine 1	BRIAR Patch Biosciences	HUP1N	Mouse	1:200 for IHC/IF
anti-Protamine 2	BRIAR Patch Biosciences	HUP2B	Mouse	1:200 for IHC
Anti-TNP2 (B-2)	Santa Cruz Biotechnology	sc-393843	Mouse	1:200 for IHC
Goat anti Rabbit IgG (H + L)-HRP	Bio-rad	Cat# 1706515	-	1:5000 for WB
Goat anti-Mouse IgG (H + L)-HRP	Bio-rad	Cat# 1706516	-	1:5000 for WB
Alexa Fluor™ 568 goat Anti-mouse IgG (H+L)	Invitrogen	Cat# A11004	-	1:500 for IF
Alexa Fluor™ 488 goat Anti-rabbit IgG (H+L)	Invitrogen	Cat# A11034	-	1:500 for IF

WB: Western Blot Analysis, IF: Immunofluorescence and IHC: Immunohistochemistry.

Table S1. Primary Antibodies.

Products	Company	Reference
Bovine Serum Albumin (BSA)	Euromedex	# 04-100-812-C
Clarity™ Western ECL Substrate	Bio-Rad	Cat # 170-5061
Clarity Max™ Western ECL Substrate	Bio-Rad	Cat # 170-5062
Complete Protease inhibitor EDTA-free	Sigma-Aldrich	# 4693159001
Dako Fluorescent Mounting Medium	Dako	S3023
DL-Dithiothreitol (DTT)	Euromedex	EU0006
DNase I recombinant	Roche	4536282001
DPBS	Gibco	14200-075
Ethylenediaminetetraacetic acid disodium salt dihydrate (EDTA)	Sigma-Aldrich	E6635
Ethylene glycol-bis (2-aminoethylether)-N,N,N',N'-tetraacetic acid (EGTA)	Sigma-Aldrich	E3889
Formalin solution, neutral buffered, 10%	Sigma-Aldrich	HT5012
Glutaraldehyde solution	Sigma-Aldrich	G5882
Heparin sodium salt from porcine intestinal mucosa	Sigma-Aldrich	H3149
Hoechst	Invitrogen	# 33342
Human Chorionic Gonadotropin (HCG)	MED-VET	Chorulon 1500 UI
Laemmli Sample Buffer 4x	BioRad	# 1610747
Magnesium Chloride (MgCl ₂)	Euromedex	2189
Mayer's Hematoxylin Solution	Sigma-Aldrich	MHS16
M2 medium	Sigma-Aldrich	M7167
M16 medium	Sigma-Aldrich	M7292
Nitrocellulose Blotting membrane	Amersham™ Protran®	10600015
Nonidet™ P 40 Substitute (NP 40)	Sigma-Aldrich	74385
NuPAGE™ 4-12% Bis-Tris Gel	Invitrogen	NP0335BOX

Table S2. Reagents / Products used.

Potassium Ferricyanide	Sigma-Aldrich	P-8131
Potassium Ferrocyanide	Sigma-Aldrich	P-9387
Pregnant Mare Serum Gonadotropin (PMSG)	MED-VET	Syncro-part PMSG 600 UI
Saccharose	Sigma-Aldrich	S7903
Saponin from quillaja bark	Sigma-Aldrich	S7900
Sodium deoxycholate (NaDoc)	Sigma-Aldrich	D6750
Superfrost®slides	Thermo Scientific™	LCSF05
TMB ELISA Substrate (High Sensitivity)	Abcam	AB171523
Trans-Blot Turbo 5X Transfer Buffer	BioRad	Cat # 10026938
Triton™ X-100	Sigma-Aldrich	T9284
TRIzol™	Thermo Scientific™	# 15596026
TWEEN® 20	Sigma-Aldrich	P7949
Urea	Sigma-Aldrich	U5378
X-Gal	Invitrogen	B1690
96 Well Round (U) Bottom Plate	Thermo Scientific™	# 163320
450 nm Stop Solution for TMB Substrate	Abcam	AB171529

Table S2. (continued)

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Additional information

Manuscript preparation

Parts of the text have been refined, with the help of free online tools (ChatGPT and DeepL).

Author contributions

AL: experimentation, data interpretation, writing and preparation of the manuscript, FB: supervision, experimentation, data interpretation, writing and preparation of the manuscript, management of acquisition of *Atad2* KO mouse model. SB: experimentation and the maintain of *Atad2* mouse model, EL and CA *in vitro* fertilization essays. FC and EBF transcriptomic data analyses. DePu and CG: transcriptomic data generation, SR: founding acquisition and transcriptomic analysis. DaPe and AV: experimentation and data generation and analyses. SK: conception, founding acquisition and supervision, and coordination of the whole project and writing of the manuscript.

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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors analyzed the expression of ATAD2 protein in post-meiotic stages and characterized the localization of various testis-specific proteins in the testis of the Atad2 knockout (KO). By cytological analysis as well as the ATAC sequencing, the study showed that increased levels of HIRA histone chaperone, accumulation of histone H3.3 on post-meiotic nuclei, defective chromatin accessibility and also delayed deposition of protamines. Sperm from the Atad2 KO mice reduces the success of in vitro fertilization. The work was performed well, and most of the results are convincing. However, this manuscript does not suggest a molecular mechanism for how ATAD2 promotes the formation of testis-specific chromatin.

Strengths:

The paper describes the role of ATAD2 AAA+ ATPase in the proper localization of sperm-specific chromatin proteins such as protamine, suggesting the importance of the DNA replication-independent histone exchanges with the HIRA-histone H3.3 axis.

Weaknesses:

The work was performed well, and most of the results are convincing. However, this manuscript does not suggest a molecular mechanism for how ATAD2 promotes the formation of testis-specific chromatin.

<https://doi.org/10.7554/eLife.107582.2.sa3>

Reviewer #2 (Public review):

Summary:

This manuscript by Liakopoulou et al. presents a comprehensive investigation into the role of ATAD2 in regulating chromatin dynamics during spermatogenesis. The authors elegantly demonstrate that ATAD2, via its control of histone chaperone HIRA turnover, ensures proper H3.3 localization, chromatin accessibility, and histone-to-protamine transition in post-meiotic male germ cells. Using a new well-characterized Atad2 KO mouse model, they show that ATAD2 deficiency disrupts HIRA dynamics, leading to aberrant H3.3 deposition, impaired transcriptional regulation, delayed protamine assembly, and defective sperm genome compaction. The study bridges ATAD2's conserved functions in embryonic stem cells and cancer to spermatogenesis, revealing a novel layer of epigenetic regulation critical for male fertility.

Strengths:

The MS first demonstration of ATAD2's essential role in spermatogenesis, linking its expression in haploid spermatids to histone chaperone regulation by connecting ATAD2-dependent chromatin dynamics to gene accessibility (ATAC-seq), H3.3-mediated transcription, and histone eviction. Interestingly and surprisingly, sperm chromatin defects in Atad2 KO mice impair only in vitro fertilization but not natural fertility, suggesting unknown compensatory mechanisms in vivo.

Weaknesses:

The MS is robust and there are not big weaknesses

The authors have addressed all the queries successfully.

<https://doi.org/10.7554/eLife.107582.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors generated knockout mice for Atad2, a conserved bromodomain-containing factor expressed during spermatogenesis. In Atad2 KO mice, HIRA, a chaperone for histone variant H3.3, was upregulated in round spermatids, accompanied by an apparent increase in H3.3 levels. Furthermore, the sequential incorporation and removal of TH2B and PRM1 during spermiogenesis were partially disrupted in the absence of ATAD2, possibly due to delayed histone removal. Despite these abnormalities, Atad2 KO male mice were able to produce offspring normally.

Strengths:

The manuscript addresses the biological role of ATAD2 in spermatogenesis using a knockout mouse model, providing a valuable in vivo framework to study chromatin regulation during male germ cell development. The observed redistribution of H3.3 in round spermatids is clearly presented and suggests a previously unappreciated role of ATAD2 in histone variant

dynamics. The authors also document defects in the sequential incorporation and removal of TH2B and PRM1 during spermiogenesis, providing phenotypic insight into chromatin transitions in late spermatogenic stages. Overall, the study presents a solid foundation for further mechanistic investigation into ATAD2 function.

Weaknesses:

While the manuscript reports the gross phenotype of *Atad2* KO mice, the findings remain largely superficial and do not convincingly demonstrate how ATAD2 deficiency affects chromatin.

<https://doi.org/10.7554/eLife.107582.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors analyzed the expression of ATAD2 protein in post-meiotic stages and characterized the localization of various testis-specific proteins in the testis of the Atad2 knockout (KO). By cytological analysis as well as the ATAC sequencing, the study showed that increased levels of HIRA histone chaperone, accumulation of histone H3.3 on post-meiotic nuclei, defective chromatin accessibility and also delayed deposition of protamines. Sperm from the Atad2 KO mice reduces the success of in vitro fertilization. The work was performed well, and most of the results are convincing. However, this manuscript does not suggest a molecular mechanism for how ATAD2 promotes the formation of testis-specific chromatin.

We would like to take this opportunity to highlight that the present study builds on our previously published work, which examined the function of ATAD2 in both yeast *S. pombe* and mouse embryonic stem (ES) cells (Wang et al., 2021). In yeast, using genetic analysis we showed that inactivation of HIRA rescues defective cell growth caused by the absence of ATAD2. This rescue could also be achieved by reducing histone dosage, indicating that the toxicity depends on histone over-dosage, and that HIRA toxicity, in the absence of ATAD2, is linked to this imbalance.

Furthermore, HIRA ChIP-seq performed in mouse ES cells revealed increased nucleosome-bound HIRA, particularly around transcription start sites (TSS) of active genes, along with the appearance of HIRA-bound nucleosomes within normally nucleosome-free regions (NFRs). These findings pointed to ATAD2 as a major factor responsible for unloading HIRA from nucleosomes. This unloading function may also apply to other histone chaperones, such as FACT (see Wang et al., 2021, Fig. 4C).

In the present study, our investigations converge on the same ATAD2 function in the context of a physiologically integrated mammalian system—spermatogenesis. Indeed, in the absence of ATAD2, we observed H3.3 accumulation and enhanced H3.3-mediated gene expression. Consistent with this functional model of ATAD2— unloading chaperones from histone- and non-histone-bound chromatin—we also observed defects in histone-toprotamine replacement.

Together, the results presented here and in Wang et al. (2021) reveal an underappreciated regulatory layer of histone chaperone activity. Previously, histone chaperones were primarily

understood as factors that load histones. Our findings demonstrate that we must also consider a previously unrecognized regulatory mechanism that controls assembled histone-bound chaperones. This key point was clearly captured and emphasized by Reviewer #2 (see below).

Strengths:

The paper describes the role of ATAD2 AAA+ ATPase in the proper localization of sperm-specific chromatin proteins such as protamine, suggesting the importance of the DNA replication-independent histone exchanges with the HIRA-histone H3.3 axis.

Weaknesses:

(1) Some results lack quantification.

We will consider all the data and add appropriate quantifications where necessary.

(2) The work was performed well, and most of the results are convincing. However, this manuscript does not suggest a molecular mechanism for how ATAD2 promotes the formation of testis-specific chromatin.

Please see our comments above.

Reviewer #2 (Public review):

Summary:

This manuscript by Liakopoulou et al. presents a comprehensive investigation into the role of ATAD2 in regulating chromatin dynamics during spermatogenesis. The authors elegantly demonstrate that ATAD2, via its control of histone chaperone HIRA turnover, ensures proper H3.3 localization, chromatin accessibility, and histone-toprotamine transition in post-meiotic male germ cells. Using a new well-characterized Atad2 KO mouse model, they show that ATAD2 deficiency disrupts HIRA dynamics, leading to aberrant H3.3 deposition, impaired transcriptional regulation, delayed protamine assembly, and defective sperm genome compaction. The study bridges ATAD2's conserved functions in embryonic stem cells and cancer to spermatogenesis, revealing a novel layer of epigenetic regulation critical for male fertility.

Strengths:

The MS first demonstration of ATAD2's essential role in spermatogenesis, linking its expression in haploid spermatids to histone chaperone regulation by connecting ATAD2-dependent chromatin dynamics to gene accessibility (ATAC-seq), H3.3-mediated transcription, and histone eviction. Interestingly and surprisingly, sperm chromatin defects in Atad2 KO mice impair only in vitro fertilization but not natural fertility, suggesting unknown compensatory mechanisms in vivo.

Weaknesses:

The MS is robust and there are not big weaknesses

Reviewer #3 (Public review):

Summary:

The authors generated knockout mice for Atad2, a conserved bromodomain-containing factor expressed during spermatogenesis. In Atad2 KO mice, HIRA, a chaperone for histone variant H3.3, was upregulated in round spermatids, accompanied by an apparent increase in H3.3 levels. Furthermore, the sequential incorporation and removal

of TH2B and PRM1 during spermiogenesis were partially disrupted in the absence of ATAD2, possibly due to delayed histone removal. Despite these abnormalities, Atad2 KO male mice were able to produce offspring normally.

Strengths:

The manuscript addresses the biological role of ATAD2 in spermatogenesis using a knockout mouse model, providing a valuable in vivo framework to study chromatin regulation during male germ cell development. The observed redistribution of H3.3 in round spermatids is clearly presented and suggests a previously unappreciated role of ATAD2 in histone variant dynamics. The authors also document defects in the sequential incorporation and removal of TH2B and PRM1 during spermiogenesis, providing phenotypic insight into chromatin transitions in late spermatogenic stages. Overall, the study presents a solid foundation for further mechanistic investigation into ATAD2 function.

Weaknesses:

While the manuscript reports the gross phenotype of Atad2 KO mice, the findings remain largely superficial and do not convincingly demonstrate how ATAD2 deficiency affects chromatin dynamics. Moreover, the phenotype appears too mild to elucidate the functional significance of ATAD2 during spermatogenesis.

We respectfully disagree with the statement that our findings are largely superficial. Based on our investigations of this factor over the years, it has become evident that ATAD2 functions as an auxiliary factor that facilitates mechanisms controlling chromatin dynamics (see, for example, Morozumi et al., 2015). These mechanisms can still occur in the absence of ATAD2, but with reduced efficiency, which explains the mild phenotype we observed.

This function, while not essential, is nonetheless an integral part of the cell's molecular biology and should be studied and brought to the attention of the broader biological community, just as we study essential factors. Unfortunately, the field has tended to focus primarily on core functional actors, often overlooking auxiliary factors. As a result, our decade-long investigations into the subtle yet important roles of ATAD2 have repeatedly been met with skepticism regarding its functional significance, which has in turn influenced editorial decisions.

We chose eLife as the venue for this work specifically to avoid such editorial barriers and to emphasize that facilitators of essential functions do exist. They deserve to be investigated, and the underlying molecular regulatory mechanisms must be understood.

(1) Figures 4-5: The analyses of differential gene expression and chromatin organization should be more comprehensive. First, Venn diagrams comparing the sets of significantly differentially expressed genes between this study and previous work should be shown for each developmental stage. Second, given the established role of H3.3 in MSCI, the effect of Atad2 knockout on sex chromosome gene expression should be analyzed. Third, integrated analysis of RNA-seq and ATAC-seq data is needed to evaluate how ATAD2 loss affects gene expression. Finally, H3.3 ChIP-seq should be performed to directly assess changes in H3.3 distribution following Atad2 knockout.

(1) In the revised version, we will include Venn diagrams to illustrate the overlap in significantly differentially expressed genes between this study and previous work. However, we believe that the GSEAs presented here provide stronger evidence, as they indicate the statistical significance of this overlap (p-values). In our case, we observed p-value < 0.01 (**) and p < 0.001 (***).

(2) Sex chromosome gene expression was analyzed and is presented in Fig. 5C.

(3) The effect of ATAD2 loss on gene expression is shown in Fig. 4A, B, and C as histograms, with statistical significance indicated in the middle panels.

(4) Although mapping H3.3 incorporation across the genome in wild-type and Atad2 KO cells would have been informative, the available anti-H3.3 antibody did not work for ChIP-seq, at least in our hands. The authors of Fontaine et al., 2022, who studied H3.3 during spermatogenesis in mice, must have encountered the same problem, since they tagged the endogenous H3.3 gene to perform their ChIP experiments.

(2) Figure 3: The altered distribution of H3.3 is compelling. This raises the possibility that histone marks associated with H3.3 may also be affected, although this has not been investigated. It would therefore be important to examine the distribution of histone modifications typically associated with H3.3. If any alterations are observed, ChIP-seq analyses should be performed to explore them further.

Based on our understanding of ATAD2's function—specifically its role in releasing chromatin-bound HIRA—in the absence of ATAD2 the residence time of both HIRA and H3.3 on chromatin increases. This results in the detection of H3.3 not only on sex chromosomes but across the genome. Our data provide clear evidence of this phenomenon. The reviewer is correct in suggesting that the accumulated H3.3 would carry H3.3-associated histone PTMs; however, we are unsure what additional insights could be gained by further demonstrating this point.

(3) Figure 7: While the authors suggest that pre-PRM2 processing is impaired in Atad2 KO, no direct evidence is provided. It is essential to conduct acid-urea polyacrylamide gel electrophoresis (AU-PAGE) followed by western blotting, or a comparable experiment, to substantiate this claim.

Figure 7 does not suggest that pre-PRM2 processing is affected in Atad2 KO; rather, this figure—particularly Fig. 7B—specifically demonstrates that pre-PRM2 processing is impaired, as shown using an antibody that recognizes the processed portion of pre-PRM2. ELISA was used to provide a more quantitative assessment; however, in the revised manuscript we will also include a western blot image.

(4) HIRA and ATAD2: Does the upregulation of HIRA fully account for the phenotypes observed in Atad2 KO? If so, would overexpression of HIRA alone be sufficient to phenocopy the Atad2 KO phenotype? Alternatively, would partial reduction of HIRA (e.g., through heterozygous deletion) in the Atad2 KO background be sufficient to rescue the phenotype?

These are interesting experiments that require the creation of appropriate mouse models, which are not currently available.

(5) The mechanism by which ATAD2 regulates HIRA turnover on chromatin and the deposition of H3.3 remains unclear from the manuscript and warrants further investigation.

The Reviewer is absolutely correct. In addition to the points addressed in response to Reviewer #1's general comments (see above), it would indeed have been very interesting to test the segregase activity of ATAD2 (likely driven by its AAA ATPase activity) through in vitro experiments using the *Xenopus* egg extract system described by Tagami et al., 2004. This system can be applied both in the presence and absence (via immunodepletion) of ATAD2 and would also allow the use of ATAD2 mutants, particularly those with inactive AAA ATPase or bromodomains. However, such experiments go well beyond the scope of this study, which focuses on the role of ATAD2 in chromatin dynamics during spermatogenesis.

References:

- (1) Wang T, Perazza D, Boussouar F, Cattaneo M, Bougdour A, Chuffart F, Barral S, Vargas A, Liakopoulou A, Puthier D, Bargier L, Morozumi Y, Jamshidikia M, Garcia-Saez I, Petosa C, Rousseaux S, Verdel A, Khochbin S. ATAD2 controls chromatin-bound HIRA turnover. *Life Sci Alliance*. 2021 Sep 27;4(12):e202101151. doi: 10.26508/lsa.202101151. PMID: 34580178; PMCID: PMC8500222.
- (2) Morozumi Y, Boussouar F, Tan M, Chaikuad A, Jamshidikia M, Colak G, He H, Nie L, Petosa C, de Dieuleveult M, Curtet S, Vitte AL, Rabatel C, Debernardi A, Cosset FL, Verhoeyen E, Emadali A, Schweifer N, Gianni D, Gut M, Guardiola P, Rousseaux S, Gérard M, Knapp S, Zhao Y, Khochbin S. Atad2 is a generalist facilitator of chromatin dynamics in embryonic stem cells. *J Mol Cell Biol*. 2016 Aug;8(4):349-62. doi: 10.1093/jmcb/mjv060. Epub 2015 Oct 12. PMID: 26459632; PMCID: PMC4991664.
- (3) Fontaine E, Papin C, Martinez G, Le Gras S, Nahed RA, Héry P, Buchou T, Ouarrarhni K, Favier B, Gautier T, Sabir JSM, Gerard M, Bednar J, Arnoult C, Dimitrov S, Hamiche A. Dual role of histone variant H3.3B in spermatogenesis: positive regulation of piRNA transcription and implication in X-chromosome inactivation. *Nucleic Acids Res*. 2022 Jul 22;50(13):7350-7366. doi: 10.1093/nar/gkac541. PMID: 35766398; PMCID: PMC9303386.
- (4) Tagami H, Ray-Gallet D, Almouzni G, Nakatani Y. Histone H3.1 and H3.3 complexes mediate nucleosome assembly pathways dependent or independent of DNA synthesis. *Cell*. 2004 Jan 9;116(1):51-61. doi: 10.1016/s0092-8674(03)01064-x. PMID: 14718166.

Recommendations for the authors:

Reviewing Editor Comments:

I note that the reviewers had mixed opinions about the strength of the evidence in the manuscript. A revision that addresses these points would be welcome.

Reviewer #1 (Recommendations for the authors):

Major points:

(1) No line numbers: It is hard to point out the issues.

The revised version harbors line numbers.

(2) Given the results shown in Figure 3 and Figure 4, it is nice to show the chromosomal localization of histone H3.3 in spermatocytes or post-meiotic cells by Chromatin-immunoprecipitation sequencing (ChIP-seq).

Although mapping H3.3 incorporation across the genome in wild-type and Atad2 KO cells would have been informative, the available anti-H3.3 antibody did not work for ChIP-seq in our hands. In fact, this antibody is not well regarded for ChIP-seq. For example, Fontaine et al. (2022), who investigated H3.3 during spermatogenesis in mice, circumvented this issue by tagging the endogenous H3.3 genes for their ChIP experiments.

(3) Figure 7B and 8: Why the authors used ELISA for the protein quantification. At least, western blotting should be shown.

ELISA is a more quantitative method than traditional immunoblotting. Nevertheless, as requested by the reviewer, we have now included a corresponding western blot in Fig. S3.

(4) For readers, please add a schematic pathway of histone-protamine replacement in sperm formation in Fig.1 and it would be nice to have a model figure, which contains the

authors' idea in the last figure.

As requested by this reviewer, we have now included a schematic model in Figure 9 to summarize the main conclusions of our work.

Minor points:

(1) Page 2, the second paragraph, "pre-PRM2: Please explain more about pre-PRM2 and/or PRM2 as well as PRM1 (Figure 6).

More detailed descriptions of PRM2 processing are now given in this paragraph.

(2) Page 3, bottom paragraph, line 1: "KO" should be "knockout (KO)".

Done.

(3) Page 4, second paragraph bottom: Please explain more about the protein structure of germ-line-specific ATAD2S: how it is different from ATAD2L. Germ-line specific means it is also expressed in ovary?

As Atad2 is predominantly expressed in embryonic stem cells and in spermatogenic cells, we replaced all through the text germ-line specific by more appropriate terms.

(4) Figure 1C, western blotting: Wild-type testis extracts, both ATAD2L and -S are present. Does this mean that ATAD2L is expressed in both germ line as well as supporting cells. Please clarify this and, if possible, show the western blotting of spermatids well as spermatocytes.

Figure 1D shows sections of seminiferous tubules from Atad2 KO mice, in which lacZ expression is driven by the endogenous Atad2 promoter. The results indicate that Atad2 is expressed mainly in post-meiotic cells. Most labeled cells are located near the lumen, whereas the supporting Sertoli cells remain unlabeled. Sertoli cells, which are anchored to the basal lamina, span the entire thickness of the germinal epithelium from the basal lamina to the lumen. Their nuclei, however, are usually positioned closer to the basal membrane. Thus, the observed lacZ expression pattern argues against substantial Atad2 expression in Sertoli cells.

(5) Figure 1C: Please explain a bit more about the reduction of ATAD2 proteins in heterozygous mice.

Done

(6) Figure 1C: Genotypes of the mice should be shown in the legend.

Done

(7) Figure 1D: Please add a more magnified image of the sections to see the staining pattern in the seminiferous tubules.

The magnification does not bring more information since we lose the structure of cells within tubules due the nature of treatment of the sections for X-gal staining. Please see comments to question 1C to reviewer 2

(8) Page 5, first paragraph, line 2, histone dosage: What do the authors meant by the histone dosage? Please explain more or use more appropriate word.

"Histone dosage" refers to the amount or relative abundance of histone proteins in a cell.

(9) Figure 2A: Figure 2A: Given the result in Figure 1C, it is interesting to check the amount of HIRA in Atad2 heterozygous mice.

In Atad2 heterozygous mice, we would expect an increase in HIRA, but only to about half the level seen in the Atad2 homozygous knockout shown in Figure 2A, which is relatively modest. Therefore, we doubt that detecting such a small change—approximately half of that in Figure 2A—would yield clear or definitive results.

(10) Figure 2A, legend (n=5): What does this "n" mean? The extract of testes from "5" male mice like Figure 2B. Or 5 independent experiments. If the latter is true, it is important to share the other results in the Supplements.

"n" refers to five WT and five Atad2 KO males. The legend has been clarified as suggested by the reviewer.

(11) Figure 2A, legend, line 2, Atad2: This should be italicized.

Done

(12) Figure 2B: Please show the quantification of amounts of HIRA protein like Fig. 2A.

As indicated in the legend, what is shown is a pool of testes from 3 individuals per genotype.

(13) Figure 2B shows an increased level of HIRA in Atad2 KO testis. This suggests the role of ATAD2 in the protein degradation of HIRA. This possibility should be mentioned or tested since ATAD2 is an AAA+ ATPase.

The extensive literature on ATAD2 provides no indication that it is involved in protein degradation. In our early work on ATAD2 in the 2000s, we hypothesized that, as a member of the AAA ATPase family, ATAD2 might associate with the 19S proteasome subunit (through multimerization with the other AAA ATPase member of this regulatory subunit). However, both our published pilot studies (Caron et al., PMID: 20581866) and subsequent unpublished work ruled out this possibility. Instead, since the amount of nucleosome-bound HIRA increases in the absence of ATAD2, we propose that chromatin-bound HIRA is more stable than soluble HIRA once it has been released from chromatin by ATAD2.

(14) Page 6, second paragraph, line 5, ko: KO should be capitalized.

Done

(15) Page 6, second paragraph, line 2 from the bottom, chromatin dynamics: Throughout the text, the authors used "chromatin dynamics". However, all the authors analyzed in the current study is the localization of chromatin protein. So, it is much easier to explain the results by using "chromatin status," etc. In this context, "accessibility" is better.

We changed the term "chromatin dynamics" into a more precise term according to the context used all through the text.

(16) Figure 3: Please provide the quantification of signals of histone H3.3 in a nucleus or nuclear cytoplasm.

This request is not clear to us since we do not observe any H3.3 signal in the cytoplasm.

(17) Figure 3: As the control of specificity in post-meiotic cells, please show the image and quantification of the H3.3 signals in spermatocyte, for example.

This request is not clear to us. What specificity is meant?

(18) Figure 3, bottom panels: Please show what the white lines indicate?

The white lines indicate the limit of cell nucleus and estimated by Hoechst staining. This is now indicated in the legend of the figure.

(19) Figure 4A: Please explain more about what kind of data is here. Is this wild-type and/or Atad2 KO? The label of the Y-axis should be "mean expression level". What is the standard deviation (SD) here on the X-axis. Moreover, there is only one red open circle, but the number of this class is 5611. All 5611 genes in this group show NO expression. Please explain more.

The plot displays the mean expression levels (y-axis, labeled as "mean expression level") versus the corresponding standard deviations (x-axis), both calculated from three independent biological replicates of isolated round spermatids (Atad2 wild-type and Atad2 KO). The standard deviation reflects the variability of gene expression across biological replicates. Genes were grouped into four categories (grp1: blue, grp2: cyan, grp3: green, grp4: orange) according to the quartile of their mean expression. For grp4, all genes have no detectable expression, resulting in a mean expression of zero and a standard deviation of zero; consequently, the 5611 genes in this group are represented by a single overlapping point (red open circle) at the origin.

(20) Figure 4C: If possible, it would be better to have a statistical comparison between wild-type and the KO.

The mean profiles are displayed together with their variability (± 2 s.e.m.) across the four replicates for both ATAD2 WT (blue) and ATAD2 KO (red). For groups 1, 2, and 3, the envelopes of the curves remain clearly separated around the peak, indicating a consistent difference in signal between the two conditions. In contrast, group 4 does not present a strong signal and, accordingly, no marked difference is observed between WT and KO in this group.

(21) Figure 5, GSEA panels: Please explain more about what the GSEA is in the legend. The legend has been updated as follows:

(A) Expression profiles of post-meiotic H3.3-activated genes. The heatmap (left panel) displays the normalized expression levels of genes identified by Fontaine and colleagues as upregulated in the absence of histone H3.3 (Fontaine et al. 2022) for Atad2 WT (WT) and Atad2 KO (KO) samples at days 20, 22, 24, and 26 PP (D20 to D26). The colour scale represents the z-score of log-transformed DESeq2-normalized counts. The middle panel box plots display, pooled, normalized expression levels, aggregated across replicates and genes, for each condition (WT and KO) and each time point (D20 to D26). Statistical significance between WT and KO conditions was determined using a two-sided t-test, with p-values indicated as follows: * for p-value<0.05, ** for p-value<0.01 and *** for p-value<0.001. The right panel shows the results of gene set enrichment analysis (GSEA), which assesses whether predefined groups of genes show statistically significant differences between conditions. Here, the post-meiotic H3.3-activated genes set, identified by Fontaine et al. (2022), is significantly enriched in Atad2 KO compared with WT samples at day 26 ($p < 0.05$, FDR < 0.25). Coloured vertical bars indicate the "leading edge" genes (i.e., those contributing most to the enrichment signal), located before the point of maximum enrichment score. (B) As shown in (A) but for the "post-meiotic H3.3-repressed genes" gene set. (C) As shown in (A) but for the "sex chromosome-linked genes" gene set.

(22) Figure 6. In the KO, the number of green cells is more than red and yellow cells, suggesting the delayed maturation of green (TH2B-positive) cells. It is essential to count the number of each cell and show the quantification.

The green cells correspond to those expressing TH2B but lacking transition proteins (TP) and protamine 1 (Prm1), indicating that they are at earlier stages than elongating-condensing spermatids. Counting these green cells simply reflects the ratio of elongating/condensing spermatids to earlier-stage cells, which varies depending on the field examined. The key point in this experiment is that in wild-type mice, only red cells (elongating/condensing

spermatids) and green cells (earlier stages) are observed. By contrast, in Atad2 KO testes, a significant proportion of yellow cells appears, which are never seen in wild-type tissue. The crucial metric here is the percentage of yellow cells relative to the total number of elongating/condensing spermatids (red cells). In wild-type testes, this value is consistently 0%, whereas in Atad2 KO testes it always ranges between 50% and 100% across all fields containing substantial numbers of elongating/condensing spermatids.

(23) Figure 8A: Please show the images of sperm (heads) in the KO mice with or without decompaction.

The requested image is now displayed in Figure S5.

(24) Figure 8C: In the legend, it says $n=5$. However, there are more than 5 plots on the graph. Please explain the experiment more in detail.

The experiment is now better explained in the legend of this Figure.

Reviewer #2 (Recommendations for the authors):

While the study is rigorous and well performed, the following minor points could be addressed to strengthen the manuscript:

Figure 1C should indicate each of the different types of cells present in the sections. It would be of interest to show specifically the different post-meiotic germ cells.

With this type of sample preparation, it is difficult to precisely distinguish the different cell types within the sections. Nevertheless, the staining pattern strongly indicates that most of the intensely stained cells are post-meiotic, situated near the tubule lumens and extending roughly halfway toward the basal membrane.

In the absence of functional ATAD2, the accumulation of HIRA primarily occurs in round spermatids (Fig. 2B). If technically possible, it would be of great interest to show this by IHC of testis section.

Unfortunately, our antibody did not satisfactorily work in IHC.

The increased of H3.3 signal in Atad2 KO spermatids (Fig. 3) is interpreted because of a reduced turnover. However, alternative explanations (e.g., H3.3 misincorporation or altered chaperone affinity) should not be ruled out.

The referee is correct that alternative explanations are possible. However, based on our previous work (Wang et al., 2021; PMID: 34580178), we demonstrated that in the absence of ATAD2, there is reduced turnover of HIRAbound nucleosomes, as well as reduced nucleosome turnover, evidenced by the appearance of nucleosomes in regions that are normally nucleosome-free at active gene TSSs. We have no evidence supporting any other alternative hypothesis.

In the MS the reduced accessibility at active genes (Fig. 4) is attributed to H3.3 overloading. However, global changes in histone acetylation (e.g., H4K5ac) or other remodelers in KO cells could be also consider.

In fact, we meant that histone overloading could be responsible for the altered accessibility. This has been clearly demonstrated in case of *S. cerevisiae* in the absence of Yta7 (*S. cerevisiae* ATAD2) (PMID: 25406467).

In relation with the sperm compaction assay (Fig. 8A), the DTT/heparin/Triton protocol may not fully reflect physiological decompaction. This could be validated with alternative methods (e.g., MNase sensitivity).

The referee is right, but since this is a subtle effect as it can be judged by normal fertility, we doubt that milder approaches could reveal significant differences between wildtype and Atad2 KO sperms.

It is surprising that despite the observed alterations in the genome organization of the sperm, the natural fertility of the KO mice is not affected (Fig. 8C). This warrants deeper discussion: Is functional compensation occurring (e.g., by p97/VCP)? Analysis of epididymal sperm maturation or uterine environment could provide insights.

As detailed in the Discussion section, this work, together with our previous study (Wang et al., 2021; PMID: 34580178), highlights an overlooked level of regulation in histone chaperone activity: the release of chromatinbound factors following their interaction with chromatin. This is an energy-dependent process, driven by ATP and the associated ATPase activity of these factors. Such activity could be mediated by various proteins, such as p97/VCP or DNAJC9–HSP70, as discussed in the manuscript, or by yet unidentified factors. However, most of these mechanisms are likely to occur during the extensive histone-to-histone variant exchanges of meiosis and post-meiotic stages. To the best of our knowledge, epididymal sperm maturation and the uterine environment do not involve substantial histone-to-histone or histone-to-protamine exchanges.

The authors showed that MSCI genes present an enhancement of repression in the absence of ATAD2 by enhancing H3.3 function. It would be also of interest to analyze the behavior of the Sex body during its silencing (zygotene to pachytene) by looking at different markers (i.e., gamma-H2AX phosphorylation, Ubiquitylation etc).

The referee is correct that this is an interesting question. Accordingly, in our future work, we plan to examine the sex body in more detail during its silencing, using a variety of relevant markers, including those suggested by the reviewer. However, we believe that such investigations fall outside the scope of the present study, which focuses on the molecular relationship between ATAD2 and H3.3, rather than on the role of H3.3 in regulating sex body transcription. For a comprehensive analysis of this aspect, studies should primarily focus on the H3.3 mouse models reported by Fontaine and colleagues (PMID: 35766398).

Fig. 6: Co-staining of TH2B/TP1/PRM1 is convincing but would benefit from quantification (% cells with overlapping signals).

The green cells correspond to those expressing TH2B but lacking transition proteins (TP) and protamine 1 (Prm1), indicating that they are at earlier stages than elongating–condensing spermatids. Counting these green cells simply reflects the ratio of elongating/condensing spermatids to earlier-stage cells, which varies depending on the field examined. The key point is that in wild-type mice, only red cells (elongating/condensing spermatids) and green cells (earlier stages) are observed. By contrast, in Atad2 KO testes, a significant proportion of yellow cells appears, which are never seen in wild-type tissue. The crucial metric is the percentage of yellow cells relative to the total number of elongating/condensing spermatids (red cells). In wild-type testes, this value is consistently 0%, whereas in Atad2 KO testes it always ranges between 50% and 100% across all fields containing substantial numbers of elongating/condensing spermatids.

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